

Ceramics Undressed

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Submitted in partial fulfillment of the requirements for the degree of Master of Science in
Data Visualization at Parsons School of Design.

May 15, 2026

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Abstract

Ceramic vessels are among the oldest and most constant traces of human making. In an age of constant progress, it is often our differences - not our similarities - that are celebrated. In this project, I show how ceramic vessels offer an opportunity to find a thread of connection to one another across oceans and political divides, and to our ancestors from thousands of years ago. When we undress ceramic vessels - ignoring surface decoration and focusing instead on form - similarities between objects become clear.

The technology of working clay into vessels to serve basic needs like cooking, drinking, and storage has existed for over 20,000 years. Though we've moved away from a hands-on connection with the materiality of clay - today, we interact mainly with mass-produced ceramics - we still share these same needs and use ceramics in our everyday lives.

By highlighting vessels whose forms resemble one another, I aim to emphasize that we resemble one another as well. It is not my intention to minimize the historical and artistic complexity of ceramics (or people), but rather to encourage an understanding of our connectedness, our similarity to other cultures, and our humanity through this common technology and canvas of expression and utility.

The live project:

<https://okasmin-thesis-vessels.netlify.app/>

Code for the project:

<https://github.com/oliviakasmin/okasmin-parsons-thesis-2026>

Acknowledgements

I would like to sincerely thank my thesis advisors, Daniel Sauter and Thiago Hersan, for their guidance, insight, patience, and thoughtful critique throughout the development of this project. I learned a tremendous amount from both of them and am deeply grateful for the opportunity to benefit from their unique perspectives, experiences, and expertise.

I am also grateful to my classmates for their feedback and friendship throughout the thesis process.

Finally, I would like to thank my fiancée, Cooper Zelnick, for his unwavering support.

Introduction

Some of the earliest examples of ceramics in the world come from Japan's Neolithic period and date back to roughly 13,000 years ago¹². These Jōmon pottery vessels include items that are easily recognizable as bowls and vases, which are objects that humans around the globe still use and make today. While there have been many advancements in technology, culture, and humankind since Japan's Neolithic period that should be celebrated, I'd like to invite readers and viewers to take a moment to appreciate the continuity of the medium and these forms.

This project will touch on the basic process of making ceramic vessels, the anatomy of a vessel, a basic introduction to the history of the craft and art form, and an exploration into the form and function of the vessels. The written portion will focus more on the history whereas the data visualization will focus on the form for the vessels.

Before digging into any of these topics, we should establish a shared understanding of what we're looking at. Below are some definitions to consider –

ceramic - (adj) “made from clay that has been shaped and then baked until hard” ³

vessel - (n) “a container (such as a cask, bottle, kettle, cup, or bowl) for holding something” ⁴

pottery - (n) “objects that are made out of clay by hand” or “dishes, bowls, and other objects made from baked clay” ⁵

For the sake of this project, I'll be discussing pottery and ceramics interchangeably - basically, an object made out of clay that has been exposed to heat at a high enough temperature such that it hardens; mediums such as ceramicware, earthenware, porcelain, and stoneware will all be included. Vessels encompass many types of objects that are made as a container for materials or liquids, and will include objects titled “vessel” as well as other forms such as pots, vases, and amphorae. These distinctions are all detailed later in the methodology section where data collection is discussed. Some have handles, some are figurative in shape, and some have clear purposes, such as a funerary urn or a vase, and others may have been created as purely

¹ Department of Asian Art, “Jōmon Culture (ca. 10,500–ca. 300 B.C.),” in Heilbrunn Timeline of Art History (New York: The Metropolitan Museum of Art, 2000–), accessed May 15, 2026, http://www.metmuseum.org/toah/hd/jomo/hd_jomo.htm

² Tokyo National Museum, “Jōmon: 10,000 Years of Prehistoric Art in Japan,” accessed April 12, 2026, https://www.tnm.jp/modules/r_free_page/index.php?id=1906&lang=en

³ Cambridge University Press, “Ceramic,” *Cambridge Dictionary*, accessed April 12, 2026, <https://dictionary.cambridge.org/dictionary/english/ceramic>

⁴ Merriam-Webster, “Vessel,” *Merriam-Webster.com Dictionary*, accessed April 12, 2026, <https://www.merriam-webster.com/dictionary/vessel>

⁵ Cambridge University Press, “Pottery,” *Cambridge Dictionary*

decorative objects or may have lost their intended purpose when separated from their original makers or owners.

This project will *not* focus on what the object is made of (i.e. which type of clay was used, such as porcelain) or how it was made. The color and surface decoration will be shown to expose the true nature of the vessels, but it will not be emphasized as deeply as the shapes themselves. Removing these factors allows one to focus on the shape of the object as opposed to more readily placing the object in history through these other characteristics. People's cultural, religious and even political beliefs can often be guessed at by observing their style and physical appearance, but if you remove all clothing and accessories we're all just humans with naked bodies; this is my goal with the vessels in this project - to reduce them to their most basic form so we can observe their similarities before layering back on the other visual cues and differences that our eyes are trained to focus on.

Through the written treatment of this thesis and the data visualization I hope to emphasize to you that despite often unsurmountable differences between peoples and cultures, when you strip back many of the surface details and observe our most basic habits and needs (as seen through ceramic vessels), in a sense, *we are all the same*.

The practice of pottery shows us how connected we really are.

Treatment

Brief Historical Introduction: Craft and Art

“But now, O LORD, thou art our father; we are the clay, and thou our potter; and we all are the work of thy hand.” (Isaiah 64:8) ⁶

The history of ceramics is uniquely tied to the story of humans. This brief historical overview aims to help situate readers, touch on the complexity and extensiveness of the history of this craft, and begin to give some context for the objects.

While it's hard to pinpoint the single earliest ceramic object that was made, historians largely agree that the technology dates back tens of thousands of years. It was long believed that prehistoric figural, sculptural objects were the first ceramics created and that functional pots weren't made until humans had settled down more away from a nomadic lifestyle⁷, but a recent discovery of a fragment from a 20,000 years old bowl found in Southern China indicates otherwise⁸. It is similarly hard to identify where the technology was first utilized. In fact, the practice of making pottery developed across the globe by many different cultures, and there is no single place of invention.⁹ In Neolithic Japan, known as the Jōmon period, deep cooking containers were made out of pottery as early as 10,500 BCE¹⁰, and in Mesopotamia in the Hassuna period from 5,000 and 4,500 BCE peasants made pots including bowls and globular jars.¹¹

⁶ *The Holy Bible, King James Version*, Isaiah 64:8, accessed May 15, 2026, <https://www.kingjamesbibleonline.org/Isaiah-64-8/>

⁷ Hugo Munsterberg and Marjorie Munsterberg, *World Ceramics: From Prehistoric to Modern Times* (New York: Penguin Group, 1998), 13.

⁸ BBC News, “‘Oldest Pottery’ Found in China,” *BBC News*, June 28, 2012, <https://www.bbc.com/news/science-environment-18602281>

⁹ Hugo Munsterberg and Marjorie Munsterberg, *World Ceramics*, 13

¹⁰ Department of Asian Art, “Jōmon Culture (ca. 10,500–ca. 300 B.C.).”

¹¹ Emmanuel Cooper, *A History of Pottery* (New York: St. Martin's Press, 1972), 21.



Figure 1: *Deep Pot (Fuka-bachi)*, Japan, Middle Jomon period (ca. 3500–2500 BCE)
(<https://www.metmuseum.org/art/collection/search/44905>)

Since its beginnings, ceramics has been both a craft and an art form that has a unique treatment across cultures as well as a shared history through trade, war, cultural transmission and human development; in many ways, this history is a mirror for how people interact with one another.

A study of ceramics made by European hunter-gatherers as early as 5900 BCE reflects some of the more positive, collaborative traits of humans: “Analysis of the forms, decoration and technological choices suggests that knowledge of pottery spread through a process of cultural transmission... Taken together the evidence supports kinship-driven, super-regional communication networks that existed long before other major innovations such as agriculture, writing, urbanism or metallurgy.”¹²

On the other hand, the exchange of influences has also happened through more forceful methods. The Roman Empire established centers for creating pottery throughout their colonies as their influence expanded, building upon practices they first learned from the Greeks and building a well organized industry that spread technical knowledge to all parts of the Empire.¹³ In 1952, the Japanese invaded Korea and abducted Korean artisans and plundered their pottery to bring them back to Japan; these kidnapped potters influenced the style of Japanese pots and helped to progress their ceramics techniques.¹⁴

Tracing the full history of ceramics and the various cultural influences (friendly and otherwise) is out of scope here. However, understanding the history of pottery—and the ways in which that

¹² Ekaterina Dolbunova et al., “The Transmission of Pottery Technology among Prehistoric European Hunter-Gatherers,” *Nature Human Behaviour* 7 (2023): 171–183, <https://doi.org/10.1038/s41562-022-01491-8>

¹³ Emmanuel Cooper, *A History of Pottery*, 59–65.

¹⁴ Robbin Hopper, *Functional Pottery: Form and Aesthetic in Pots of Purpose* (Iola, WI: Krause Publications, 2000), 38–39.

history is intertwined with the history of human civilization itself—demonstrates the universality of pottery and our shared human commitment to the medium and the forms.

How Pots are Made: The Basics

“The qualities which make clay immediately attractive to children are probably qualities which attracted Stone Age man. What could be easier than to pick up a lump of clay by the river or lake and press it into shapes with the fingers?”¹⁵

While a deep understanding of how ceramics are made isn't necessary for the sake of understanding this project, I believe some sense of how these objects come to be is helpful to grasping the universality and fundamentally human nature of pottery. Since surface decoration isn't emphasized in this project, I won't touch on the many ways that this can be accomplished. This is by no means a professional report of how to make ceramics, but just the perspective of a humble hobbyist whose instruction has been based in Brooklyn, NY. While there are myriad techniques for making pottery that differ by century, by culture, and even by individual, my aim is to cover the broad strokes.

The most elemental way to make a ceramic pot is: first, source clay (perhaps directly from the earth), second, shape the clay with your hands or another basic tool, and finally, apply high heat to the clay so that it hardens (which is called *firing* the clay). Clay is a basic material that can be found in varying forms around the world - it's plastic (i.e. flexible) when wet and brittle (i.e. hard and shatterable) when dry. One might shape the clay by pinching it into a simple pot, by rolling it into coils and then stacking them upon one another, or using an existing form (perhaps a dried out gourd or a simple basket) as a mould¹⁶. Applying heat to the clay to *fire* it can be as simple as putting it directly into a hot enough fire; today we usually *fire* ceramics in a kiln. The type of clay can affect the shape as well (some types are stronger, some fire at higher temperatures resulting in different chemistry, etc.) so objects made with the same clay may share features, which can have regional implications depending on what is available.

The other main way to create vessels is by using a pottery wheel. A pottery wheel basically spins the clay around, allowing the potter to take advantage of the centrifugal force applied to shape the clay. This tool can often speed up the process and make results more consistent. Anything made on a pottery wheel will be basically round (due to the circular force), but may be altered by hand later to change the overall form or add other elements like handles.

Again, it's hard to pinpoint exactly when and where the pottery wheel first started being used, but it appears this advancement also happened across different societies. Early wheels were as

¹⁵ Emmanuel Cooper, *A History of Pottery*, 11.

¹⁶ Emmanuel Cooper, *A History of Pottery*, 16.

simple as round stones or wood that were spun by hand or foot, and electricity eventually made the spinning a little easier, but the basic concept remains the same. The axle was developed in the Middle East circa 4000-3500 BCE and later adapted for use for an early potter's wheel called the "slow wheel", which would be pushed around to allow pots to be built up more quickly and with greater uniformity.¹⁷

It's easy to understand that *how* a vessel is made will affect its final form (for example, something that's square with sharp corners probably wasn't made using a pottery wheel), but it's also interesting to consider that the reverse isn't necessarily true; by looking at the shape of a pot you can't always trace the exact steps taken to achieve that form. Research shows that potters can produce similar vessel forms through different shaping processes, reflecting variation in individual motor skills and learned practice.¹⁸ Another cross-cultural study demonstrates that when reproducing the same unfamiliar shapes, potters from different communities produce greater variation between groups than within them, indicating community-specific habits of making.¹⁹ Together, these findings suggest that while ceramic forms may appear similar across cultures, the processes that produce them are shaped by both individual skill and culturally learned techniques.

It should be noted that in modern times the production of ceramics has been industrialized and most of the ceramics we interact with haven't been made by someone's hands, but that doesn't minimize the elemental origin of this craft which essentially comes from **earth** (clay) + **water** (to make the clay plastic) + **fire** (to *fire* the clay) + some human intervention. But setting aside the industrialized ceramic production over the past century, pottery that is made by human hands is, definitionally, informed by the hands themselves. Said another way, ceramics reflect the physicality of their makers, with forms limited in size, in shape, and in structure by what can be produced by human hands and the tools we implement.

Anatomy of a Vessel

We also make pottery in relation to how it will be used. Cups, bowls, and plates are made with their weight in mind to ensure that those who use them will be able to comfortably lift them. Handles are made so that they're comfortable to hold. You might need a larger bowl to serve a larger family.

¹⁷ Emmanuel Cooper, *A History of Pottery*, 24.

¹⁸ Valentine Roux et al., "Morphological Variation of Ceramic Vessels: The Role of Shaping Processes and Individual Motor Skills," *PNAS Nexus* 3, no. 2 (2024): pgae055, <https://doi.org/10.1093/pnasnexus/pgae055>

¹⁹ Blandine Bril et al., "Cross-cultural Differences in Pottery Making: The Role of Community-specific Motor Skills," *PLOS ONE* 15, no. 9 (2020): e0239362, <https://doi.org/10.1371/journal.pone.0239362>

This fact, of course, isn't strictly a limitation. We make things that we *can* make, but also things that we *want to* make. Pottery reflects not only our physicality – our abilities and our limits – but also our lifestyles, our desires, and how we interact with the world. We make cups so that we can quench our thirst. We make bowls so that we can not only sustain ourselves, but so that we can have meals with others. We make vases so that we can live with flowers and experience beauty.

We'll explore the functional aspect of pottery in the next chapter, but it's important to note that pottery reflects not only our human ability and ingenuity, but also how we relate to the world, how we interact with the world, and even how we see ourselves.

If the pots we make are indeed reflections of us, then it's no surprise that we've long anthropomorphized them, naming their elements and parts (at least in English) after our own. Pots have lips, shoulders, and feet, and may be described as having a wide belly or a long neck. We make pottery with our hands, and the pots we make bear our fingerprints – sometimes literally. We make pottery to meet the needs of our everyday lives, and millennia later those pots survive, often as the only remaining testament to who we were and how we lived. And we name pots after ourselves, almost breathing life into them.

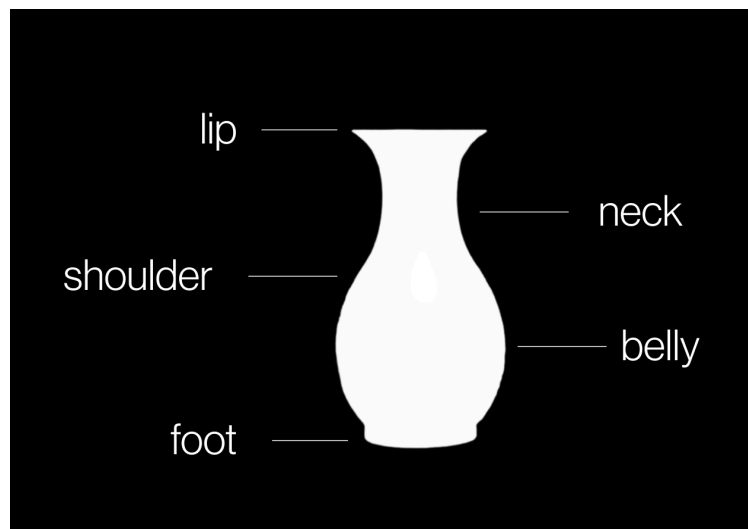


Figure 2: parts of a pot

While we shape pots, pots shape us too. Archaeologists recently discovered an ancient Greek woman with bones with unusual and asymmetrical signs of wear. When they sought to understand her unique bone structure, they realized that she was likely a master female ceramicist whose joints had been worn down by repeatedly kicking to turn her pottery wheel with the same leg and whose right side had been strengthened by leaning to one side to shape the clay, probably in service of making very large pots. She not only devoted her life to

shaping items, but she was – quite literally – shaped by her craft and by the ceramics she created.²⁰

Form and Function

The intended function of an object, by definition, affects its form. The inverse is also true. As we've seen, pottery, and specifically vessels, have long been used to improve the lives of people around the world for basic purposes like storage and food preparation.

Robbin Hopper has outlined these categories as the main functions for pottery²¹:

- Eating
- Drinking
- Storage
- Carrying
- Serving
- Cooking
- Lighting
- Planting
- Decoration
- Funereal
- Washing and perfuming

Please note that not all of these functions pertain to the vessels analyzed in my project; the specific vessels included are discussed in the methodology section below.

With each use case, and across each culture and time period, there have been differing requirements for the success of each object. For example, in climates where it's hot and water is scarce, pots for storing water historically had narrow necks to reduce evaporation.²²

Pottery was an invaluable resource for storing food in ancient times when food spoilage was a matter of life and death for families and societies. A device called a Zeer pot, which was recorded as early as 2,500 BCE in Egypt, was a method of cooling food via evaporation by placing a smaller pot inside of a larger pot and using water and sand between the two pots and

²⁰ Smithsonian Magazine, "Meet the Female Artisan Whose Masterful Ceramics Defied Ancient Greek Gender Norms," *Smithsonian Magazine*, accessed May 15, 2026, <https://www.smithsonianmag.com/smart-news/meet-female-artisan-whose-masterful-ceramics-defied-ancient-greek-gender-norms-180970288/>

²¹ Robbin Hopper, *Functional Pottery*, 10–19.

²² Robbin Hopper, *Functional Pottery*, 5.

a cloth on top to draw heat out of the innermost pot. This invention is still used today in some third-world countries with limited access to electricity.²³

The uses we developed for ceramics thousands of years ago reflected the elements of life at that time. While the tools that we use for living today might seem very different, reading this list of categories one is struck by the fact that these purposes and facets of daily life are as relevant to us as they were back then. In our era where we venerate progress and technology, and change, it's easy to forget the elements of life that are immutable; in forgetting the elements of life that are immutable it's easy to forget what connects us - pottery reminds us of that.

Common Forms

“In some oriental cultures the pot is seen as soul, where volume and form grow from the depths within, and the outside shape is a direct result of the inner form. Perhaps in the western hemisphere, we are more concerned with appearances than with truth or meaning; with clothes that obscure the form rather than glory in it.”²⁴

We have to remove the surface decoration - undress the vessels, as it were - to truly see their form without distractions. The surface decoration on ceramics is incredible, both in the artistry and the chemistry used to achieve glazes and other finishes, but this is the part that's easy to see, and easy to laud. What's harder is to focus purely on the silhouette of the object.

When we encounter people whose personal style is highly visible (bear with me for a moment while I veer heavily into stereotypes) - for example, let's say there are a few people on a subway car, one of whom is dressed in a crisp suit with a leather briefcase and one is in jeans and a graphic tee with lots of tattoos - it's natural to immediately start making assumptions about who they are, where they're from, or what communities they belong to. Do they look like us? Where are they from? What kind of event are they going to? Those visual cues shape how we interpret differences. But if you strip those markers away, what remains is a much more basic shared humanity.

This is what I do with the ceramic vessels in this project. To truly appreciate the basic shape of each object they must first be stripped bare.

There are many reasons that two vessels could look the same including:

- They're made by the same culture and follow strong stylistic norms
- One culture copied another one (knowingly or not - recall the example of the abducted Korean potters)

²³ Patrick McCarthy, “Zeer Pot: An Evaporative Cooler to Keep Food Fresh,” *Offgridweb*, February 15, 2018, <https://www.offgridweb.com/survival/zeer-pot-an-evaporative-cooler-to-keep-food-fresh/>

²⁴ Robbin Hopper, *Functional Pottery*, 9.

- It's a very basic shape that's easy and fast to make on a pottery wheel, so everyone does it
- It's a nice shape to hold in one's hands
- It's the perfect shape for its function (ie, you can pour water more easily when something has a spout, like a jug)

I'm not going to argue why any 2 shapes look the same, I just want you to consider how improbable it is that these vases, which span nearly 3000 years and 3 different continents, all resemble one another. Without a caption, could you even begin to guess the origin of the objects? It's a really basic point, but one that I think is worth making.

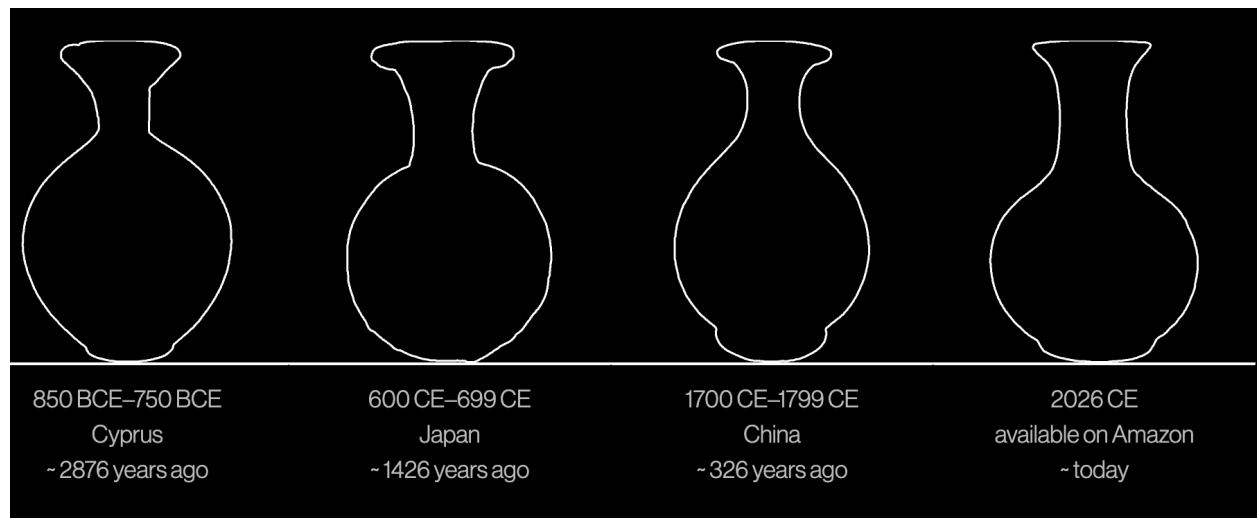


Figure 3: similarly shaped vessels; from left to right -
Jug, Cypriot, 850–750 BCE (<https://www.metmuseum.org/art/collection/search/240436>)
Long-Necked Jar (Chōkeiko), Japan, 7th century
(<https://www.metmuseum.org/art/collection/search/49359>)
Vase, China, 18th century (<https://www.metmuseum.org/art/collection/search/47427>)
Yesland Small Blue and White Porcelain Vases Set of 3, 5 Inch Tall Mini Vintage Vase
Chinoiserie Vases, Classic Decorative Bud Vases for Farmhouse Table Centerpiece, Flower
Decor (<https://www.amazon.com/dp/B0BW48XTX1>)

There are an uncountable number of different shapes for a vase, a jug, or a pot. We don't need any reminding about how different we are, and how unique, and how special, and how advanced.

We've seen that ceramics have been around basically as long as humans, that it's pretty accessible to make pottery, and that we've been using these objects for pretty similar purposes to make our lives better (or just prettier) around the world. It's hard to identify another type of physical object that has survived so long and is still being made and used today in essentially the same way. The common shapes of these objects are threads that connect us all.

Methodology

Data Collection

I constructed the dataset for this project using [The Metropolitan Museum of Art Collection API](#) as a large-scale source of ceramic vessels spanning different cultures and historical periods. I began by querying the API using the query term “vessel” and limited results to ceramic objects with publicly available images. This query returned about 40,000 objects, so I further refined my search to a more limited range of vessel-related terms, including “vase,” “bottle,” “jar,” “jug,” “ewer,” “amphora,” “urn,” “flask,” and “pitcher.” I did a few additional searches in specific departments like “Ancient West Asian Art”, “Greek and Roman Art”, and “Egyptian Art” to target some of the earliest vessels in the collection.

In part due to the necessary narrowing of search terms for my query, I intentionally excluded a number of vessel forms, including bowls, plates, and cups, to reduce the overall variation in object types. This was a curatorial choice that reflects my personal preference for the profile of a simple vase, for example, rather than a simple bowl, and my belief that the silhouettes of the vessels I included would be more visually compelling overall.

After collecting object IDs, I retrieved full object records through the API, compiled them into a unified dataset, and filtered the records to remove objects that were not suitable for analysis. I did some additional cleanup and manually reviewed both metadata and images to refine the dataset and ensure consistency across the final collection, including removing any objects that were only shards or fragments of vessels since the shapes from these images wouldn’t be visually representative of the original objects.

In addition to the query search terms, several other limitations also shaped the construction of the dataset. First, the distribution of objects across departments, cultures, and historical periods is uneven. In part, this reflects the structure of the museum’s collection itself, but it is also influenced by the historical survival of ceramic objects: newer pots are generally more likely to survive intact than the older ones.

The dataset also does not represent an exhaustive collection of ceramic vessels from The Metropolitan Museum of Art, nor was that my goal. Rather than attempting to gather every possible ceramic object, I aimed for a dataset of approximately 4,000 vessels - large enough to reveal visual patterns across cultures and time periods, while remaining computationally and logistically manageable for image processing, clustering, and analysis.

The final dataset includes **4,184 objects** across these departments:

Department	Objects	% of corpus
Asian Art	1,997	47.73
European Sculpture and Decorative Arts	690	16.49
The Michael C. Rockefeller Wing	531	12.69
Islamic Art	368	8.80
Greek and Roman Art	251	6.00
Medieval Art	128	3.06
Robert Lehman Collection	76	1.82
The American Wing	64	1.53
The Cloisters	34	0.81
Egyptian Art	32	0.76
Modern and Contemporary Art	11	0.26
Ancient West Asian Art	2	0.05
Total	4,184	100.00

Figure 4: breakdown of all objects by department

Image Processing

For each object in the dataset, I programmatically retrieved source photographs from the API. I then processed these images using [BRIA AI's RMBG-2.0](#) background-removal model, a deep-learning computer vision model designed to separate foreground objects from their backgrounds across a wide range of image types. The model generated both a transparent cutout of each vessel and a corresponding grayscale mask image representing the object's silhouette.

I tightly cropped each object to the boundaries of its mask and rescaled the images onto a standardized square canvas. Normalizing the images in this way created a consistent visual framing and resolution across the dataset so the images could be directly compared and the original scale of the object and the framing of the photograph of the object wouldn't factor into the silhouette analysis.

I then used OpenCV (Open Source Computer Vision Library), an open-source computer vision and machine learning toolkit, to process the mask images and extract contour information from each vessel. I used contour-detection methods to trace both the outer and inner contours of each vessel shape. These contours were then converted into SVG path data, producing vector-based outline representations of each vessel silhouette to use on the front end.

These outline representations became central to the project's analysis of vessel form, enabling comparisons focused on silhouette and structure.



Figure 5: image processing; from left to right -
Stem Vase with Incised and Painted Design, Thailand (Ban Chiang culture), 1000–300 BCE
(<https://www.metmuseum.org/art/collection/search/37407>)
Normalized image with background removed using BRIA
Normalized mask image from BRIA
Contour outline image from mask using OpenCV

Silhouette Feature Extraction

After generating the standardized silhouette images, I extracted a set of geometric and statistical features from each mask image in order to compare the forms. I used OpenCV to identify the primary outer silhouette as well as any interior contours, such as openings created by handles or enclosed negative space within the form. From these contours, I calculated a range of shape descriptors intended to capture different aspects of vessel geometry. These included measurements related to area, perimeter, aspect ratio, solidity, circularity, and contour extent, along with moment-based features such as Hu invariant moments and eccentricity. Together, these metrics describe qualities such as elongation, compactness, symmetry, convexity, and overall silhouette complexity.

In addition to contour statistics, I generated evenly sampled horizontal and vertical profile measurements across each silhouette. I took measurements of the x and y distances from each edge of the images to the first foreground pixel of the vessel - specifically, 64 from each edge,

adding up to 256 numbers per object. These profiles recorded normalized distances between the vessel boundary and the edges of the standardized image canvas, creating a simplified representation of the outer vessel silhouette. Aggregate measurements derived from these profiles were also used to estimate asymmetry, proportional balance, and the relationship between upper and lower vessel forms.

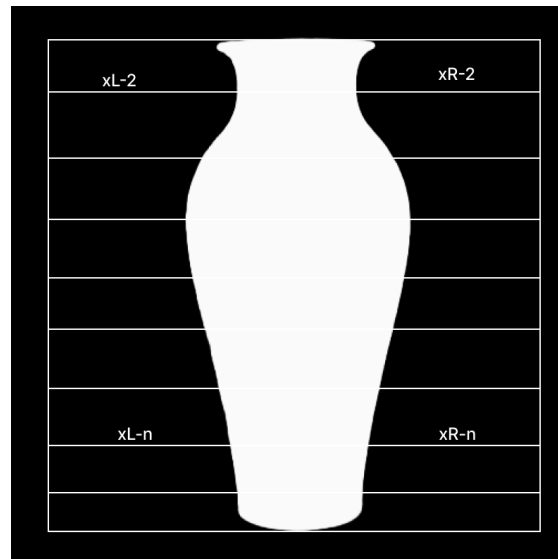


Figure 6: diagram of x measurements taken from left (xL) and right side (xR) of the image to the first pixel of the vessel

Interior contours were analyzed separately in order to capture structural features such as handles, openings, and enclosed negative space. For these interior regions, I calculated additional measurements related to area, perimeter, circularity, and proportional relationship to the main vessel body. The resulting feature set produced a numerical representation of each vessel silhouette that could be used for downstream clustering and similarity analysis.

Clustering by Silhouette Features

To group vessels according to similarities in form, I developed a clustering pipeline based on the extracted silhouette features. I organized the features into several categories including contour-based geometric descriptors, horizontal and vertical profile measurements, and interior contour features related to handles and negative space. I experimented with different weights of each feature and category to emphasize particular formal characteristics, like silhouette shape and contour geometry, and reduce the influence of less visually significant measurements.

To achieve the final clusters I combined traditional k-means clustering with a separate isolation-based approach to preserve outlier shapes that don't fit cleanly into broader formal groups. My goal was to end up with 12 clusters - a number that allowed for enough groups to represent the variation in shapes, but also wasn't too large to end up with over-clustering.

I started by separating out the outlier shapes. To identify these objects I calculated a nearest-neighbor isolation score using the weighted feature space, measuring how distant each vessel was from its closest neighbors. I separated the most isolated objects into an outlier group (cluster 0), and then repeated that process to separate out the next most isolated vessels and used k-means to group those into 2 groups, which resulted in a group of narrower and wider vessels, clusters 1 and 2. These 3 outlier groups allowed highly unusual or rare forms to remain visible within the analysis instead of being averaged into larger categories.

The remaining vessels were grouped using k-means clustering in 9 groups, producing a total of 12 final silhouette groups (including the 3 isolated outlier categories). The resulting clusters reflected recurring similarities in vessel proportions, contour structure, symmetry, and silhouette complexity across objects from different cultures and time periods.

To evaluate clustering quality, I used several standard machine-learning metrics, including cluster compactness and centroid distance, but primarily I analyzed the clusters visually by stacking the outline images of each object in the cluster to analyze the patterns.

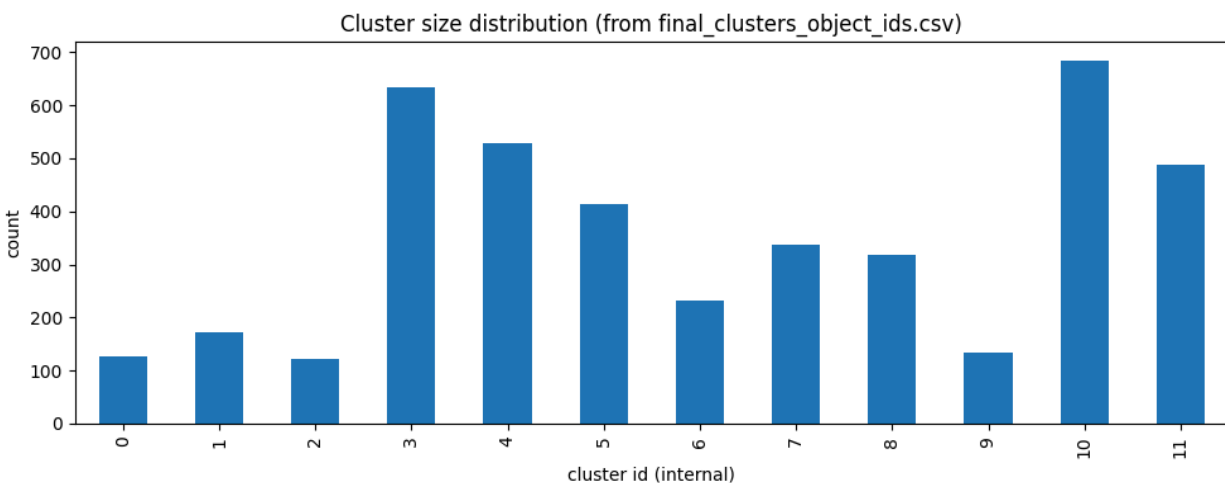


Figure 7: Bar chart of object counts per final shape cluster (cluster_0 to cluster_11). Clusters 0, 1 and 2 are the outlier clusters that are separated first and the remaining 9 are the core shape clusters.

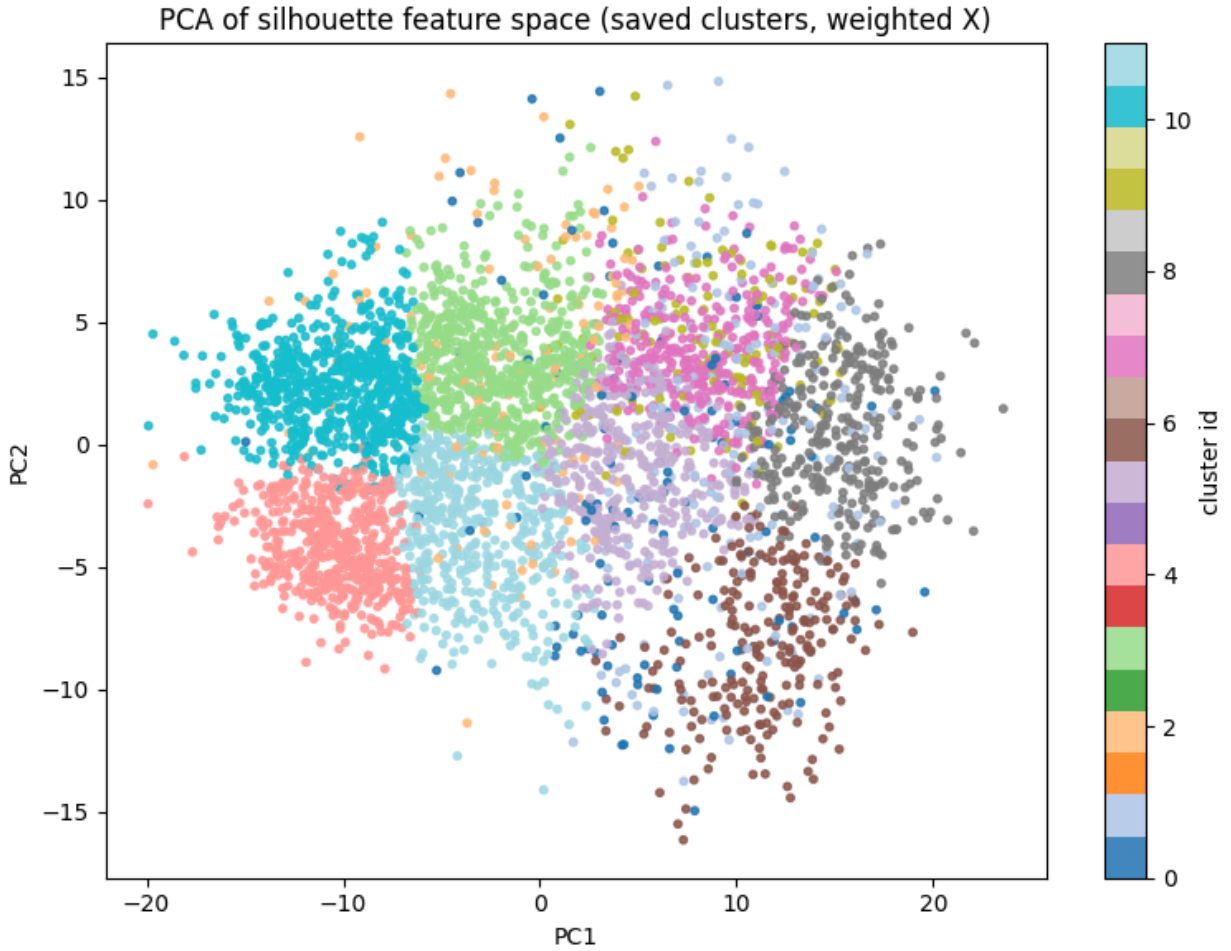


Figure 8: Two-dimensional PCA of weighted silhouette features. Each point is one vessel; color indicates final cluster assignment (cluster_0 to cluster11). The clustering pipeline used the full feature space; PCA is for visualization only.

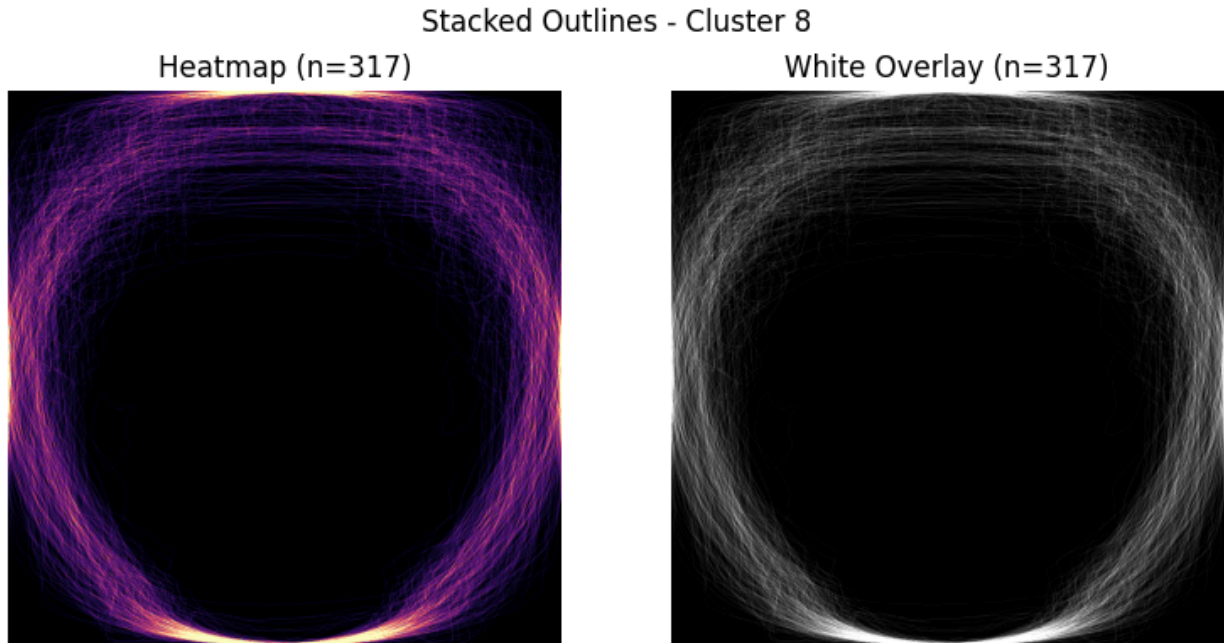


Figure 9: final heatmap analysis for cluster_8
(the images for all other clusters are in the appendix)

Shape-Based Neighbor Analysis

In addition to clustering vessels into silhouette groups, I also developed a shape-based nearest-neighbor system to group objects with similar forms across the dataset. Using the same weighted silhouette features as the clustering pipeline, I calculated pairwise Euclidean distance between all vessels to produce a ranked list of the closest shape neighbors for each object.

For every vessel, the system first identified a pool of the 20 nearest neighbors based purely on silhouette similarity. From this pool, I then selected a final set of 7 representative neighbors using additional geographic and chronological criteria. Rather than simply displaying the 7 closest objects by distance, I attempted to balance strong formal similarity with broader historical and geographic diversity to counterbalance the uneven distribution of objects in the original data collection from the museum.

To achieve this, the system prioritized neighbor combinations that included at least one object from a different geographic region than the source vessel, and favored combinations containing at least one vessel separated from the source object by at least 1000 years, and otherwise by at least 500 years.

In many instances, this methodology created fascinating groupings that really illustrated the point of the project (that the same shape is made across wide ranging time periods and cultures).

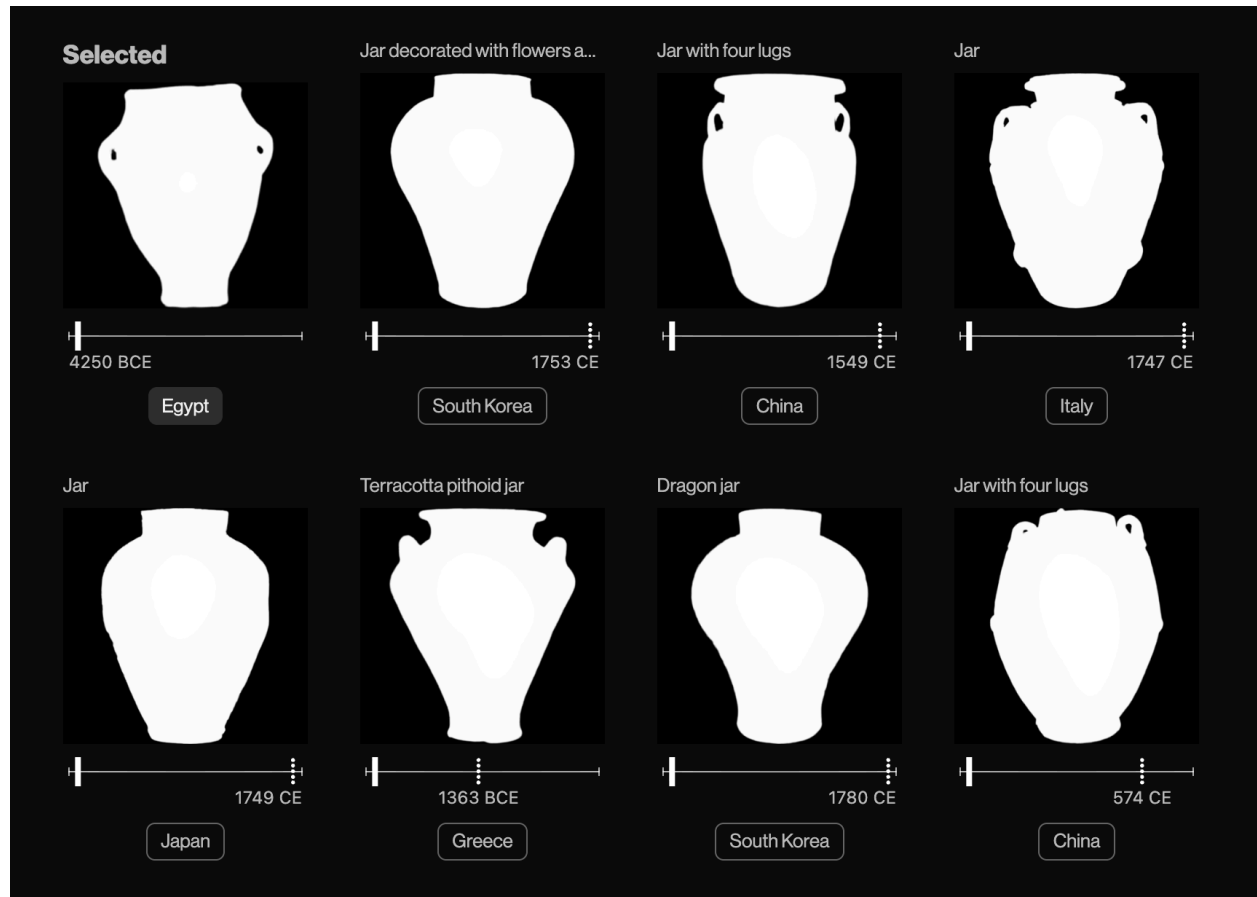


Figure 10: nearest neighbor modal shows the selected object, and the 7 chosen neighbors based on the shape, date, and place criteria.

This selected object is paired with a wide range of time periods and countries of origin

In other cases, the uneven distribution of the dataset across cultures meant that both the desired difference in time and place couldn't be achieved from only 20 neighbors. Nonetheless, computing more than 20 neighbors for each vessel seemed extreme given the size of the dataset, and by manually testing a number of shapes, 20 seemed like a good compromise (I originally started with only 10 neighbors).

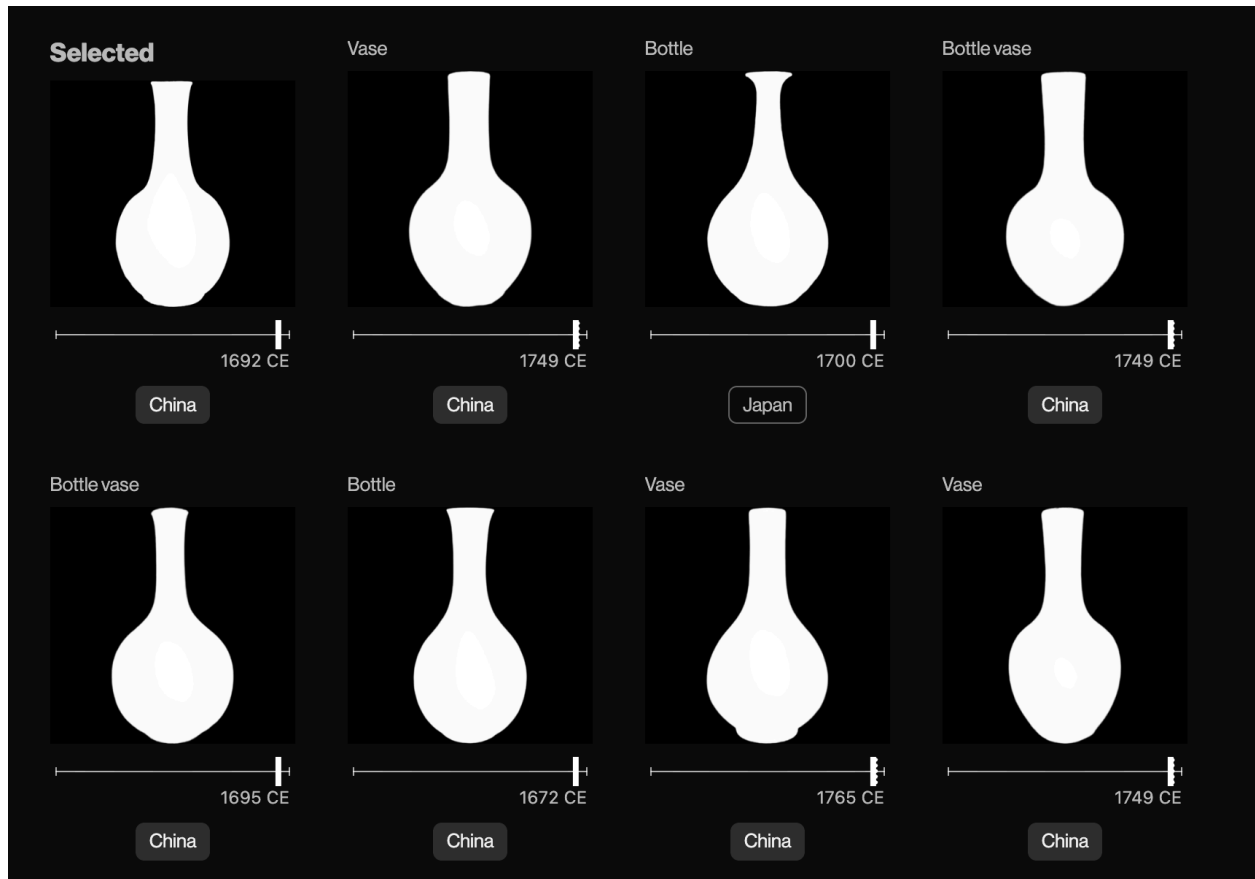


Figure 11: nearest neighbor modal shows the selected object, and the 7 chosen neighbors based on the shape, date, and place criteria.

In this case, there are so many similar vases from the same time period in China that the final 7 neighbors chosen doesn't include an object from a significantly different time, and the vase from Japan is clearly from later in the ranked neighbors since the shape is the least similar

Grouping by Use (Metadata)

To support comparing vessels by their use I started by grouping them according to their type as defined by the “objectName” and “title” fields in the metadata. Museum catalog records often use highly specific or culturally distinct object names — particularly for ancient Greek and Roman ceramics — so I created a simple mapping to object types in 10 broader groups (which are largely based on my original search terms): jar, vase, bottle, jug, pot, flask, vessel, beaker, pitcher, and amphora. For example, multiple pouring-related forms such as ewers and oinochoai were grouped under pitcher, while small oil or perfume containers such as aryballoi and lekythoi were grouped under flask.

Group by title	Count	Percent of corpus
Vase	1,279	30.57
Jar	1,025	24.50
Bottle	874	20.89
Jug	339	8.10
Pitcher	323	7.72
Flask	112	2.68
Vessel	101	2.41
Amphora	67	1.60
Pot	47	1.12
Beaker	17	0.41

Figure 12: breakdown of objects by type as determined by the title

After grouping objects by vessel type, I developed a second classification system focused on inferred use rather than formal object names alone. These “use” groupings are the ones that I ended up using the project because they’re more compelling.

While the museum metadata included titles, object names, and occasional tags, it did not provide a consistent description of how each vessel was originally used. The intended use case for an object is so dependent on its original context that it seems nearly impossible to definitely claim its purpose for many, and especially older, objects. As a result, I created a more interpretive mapping system that inferred likely use cases from the language present in object titles while attempting to avoid overly speculative assumptions.

Using this approach, I grouped vessels into several broad use categories, including pouring, storage, flask and bottle, ritual, vase, and animal-shaped. Certain titles clearly suggested a probable function — for example, ewers, pitchers, and wine vessels could reasonably be associated with pouring. Some culturally specific forms, particularly certain Ancient Greek vessel types, also strongly implied particular uses and could therefore be categorized more directly.

In other cases, however, the original function of an object was less certain. Many museum titles, particularly the broad label “vase,” reflect curatorial classification systems rather than definitive evidence of historical use. Because of this ambiguity, I chose to remain conservative in my interpretations. When I did not have enough confidence to infer a more specific use, I retained broader categories rather than imposing speculative functional labels.

The ritual category was intentionally kept small for similar reasons. Although many additional vessels may originally have served ritual, ceremonial, or funerary purposes, the available metadata was often insufficient to support those conclusions confidently. For example, I did not classify all urns as ritual objects, since urns have historically been used for a wide range of storage purposes, only some of which are related to storing human remains. Likewise, some figural or anthropomorphic vessels may have held spiritual, symbolic, or fertility-related meanings that cannot be reliably determined from the metadata alone.

I also included an animal-shaped category, which functions somewhat differently from the other use groups. These vessels highlight the recurring presence of animals, playfulness, and figural representation within ceramic traditions across cultures and time periods.

Finally, it is important to acknowledge that many objects in the dataset — particularly those produced in more recent centuries — may have been intended primarily for decorative display rather than practical use. The boundaries between functional, ceremonial, and decorative ceramics are often fluid, and this classification system should therefore be understood as an interpretive framework rather than a definitive taxonomy of historical use.

Use	Count	Percent of corpus
Storage	1,156	27.63
Vase	1,115	26.65
Pouring	944	22.56
Flask and bottle	662	15.82
Other	139	3.32
Animal-shaped	128	3.06
Ritual	40	0.96

Figure 13: breakdown of objects by possible use

Color Extraction and Grouping

While surface decoration and color are secondary to form in this project, I did want to include some color analysis at the very least to prove my point that it's hard to focus on the shape when you can see the surface variation. The process of grouping each object by color was largely manual, with some machine assistance.

Before starting to extract the color, I filtered out many black-and-white catalogue photographs while intentionally retaining objects that were themselves grayscale or monochromatic ceramics.

This distinction was necessary because the dataset included both genuinely grayscale vessels and archival photographs lacking color information.

The color extraction process presented several challenges. Museum photographs varied significantly in lighting, color correction, glare, and image quality, particularly for reflective glazed surfaces. In many cases, direct pixel analysis reduced visually rich vessels into relatively neutral average tones, especially for multicolored objects with detailed decoration. I ultimately used the Python library [Colorgram.py](#) to generate representative palettes of dominant colors for each vessel image, which was better than the averaged pixels, but still couldn't solve for the inconsistent quality of the images themselves.

I attempted to cluster the images by palette directly using k-means, and also into pre-determined groups like “blue” and “green” based on their hue, but both of these methods did not produce good results.

In the end, I used k-means clustering based on HSV and Lab color spaces for each palette to group the images and then I manually separated out the objects into about 20 groups, which I finally combined into the 12 that appear in the project. The clustering sped up this process by sometimes grouping things that were similar and allowing me to separate out larger quantities than I could have manually, but much of this process was manual, especially in handling the multicolored vessels with intricate decoration.

Color	Count	% of color-designated total (3,147)
blue	128	4.07
green	206	6.55
blue and white	305	9.69
yellow	48	1.53
red	156	4.96
multicolor	670	21.29
white	131	4.16
terracotta	480	15.25
light warm brown	220	6.99
cooler tan	299	9.50
muddier brown	283	8.99
dark	221	7.02

Figure 14: breakdown of objects into color groups

Date and Location (Metadata)

Across all entry points — shape, use, and color — I wanted viewers to be able to see how objects were distributed across both geography and time. To support this, each subgroup could also be explored through interactive timeline and map visualizations.

Most objects in the dataset were associated with a date range rather than a single year (for example, 4000–3500 BCE). In order to place these objects along a continuous timeline, I calculated a single representative date by taking the midpoint of the range. Because the dataset spans thousands of years, I then grouped objects into 500-year intervals in order to create a more readable and interpretable temporal visualization.

Date bucket	Objects	% of corpus
-4500:-4001	2	0.05
-4000:-3501	3	0.07
-3500:-3001	15	0.36
-3000:-2501	8	0.19
-2500:-2001	33	0.79
-2000:-1501	23	0.55
-1500:-1001	27	0.65
-1000:-501	147	3.51
-500:-1	255	6.09
0:499	289	6.91
500:999	310	7.41
1000:1499	565	13.50
1500:1999	2,507	59.92

Figure 15: breakdown of objects into date buckets for the timeline

Geographic metadata was highly inconsistent across departments and objects, reflecting both differing curatorial conventions and likely variations in the historical information available when objects entered the collection. Place information could appear in multiple fields — including city, region, country, culture, and artist nationality — and was often incomplete, ambiguous, and formatted differently across records.

To create a more consistent geographic dataset, I developed a normalization pipeline that combined and cleaned these metadata fields into a single best-guess location string for each object. This process included resolving culturally specific place names, standardizing nationality and regional references, reducing ambiguity where possible, and assigning confidence thresholds for whether a location was suitable for mapping.

The normalized locations were then passed to the [Mapbox Geocoding API](#) in order to generate latitude and longitude coordinates for visualization within the project's interactive map.

The final locations include 36 distinct countries.

Country	Count	% of all rows
china	1,310	31.31
japan	581	13.89
peru	443	10.59
iran	274	6.55
united kingdom	211	5.04
italy	129	3.08
greece	121	2.89
france	111	2.65
egypt	107	2.56
south korea	94	2.25
germany	64	1.53
cyprus	63	1.51
united states	63	1.51
mexico	58	1.39
syria	50	1.20
thailand	32	0.76
spain	30	0.72
iraq	20	0.48
turkey	18	0.43
vietnam	14	0.33
guatemala	11	0.26

ecuador	7	0.17
mali	6	0.14
denmark	5	0.12
netherlands	4	0.10
colombia	3	0.07
switzerland	3	0.07
french southern and antarctic lands	2	0.05
india	2	0.05
panama	2	0.05
belgium	1	0.02
czech republic	1	0.02
ireland	1	0.02
lebanon	1	0.02
russia	1	0.02
sudan	1	0.02

Figure 16: breakdown of objects into countries

Visualization Techniques & Design Decisions

The visual design of this project was shaped by the goal of emphasizing form. Throughout the interface, I used a restrained black-and-white visual language (and occasionally some gray) in order to reduce distraction from the vessel silhouette and structure. This approach reinforced the project's central idea of “undressing” ceramics by stripping away visual markers that often dominate how objects are categorized and interpreted.

To further emphasize form, I used SVG-based outline representations throughout the project. These vector outlines allowed vessel shapes to be animated and transformed fluidly across the interface. The animations were intended to evoke the plasticity of clay itself: forms appear to emerge from one another, morphing and taking shape in ways that suggest both continuity and variation.

Across many views, vessels are visually grounded on a simple horizontal “shelf” line. This design choice helped situate the objects within a shared physical space while also creating

visual consistency. The shelf motif reinforced the idea of viewing ceramics as collections of related forms rather than isolated museum artifacts.

Several visualization strategies were used throughout the project. Shape clusters were represented through stacked outline visualizations, in which many vessel silhouettes were layered together to reveal the variation within each group. These dense accumulations of outlines made it possible to visually perceive both similarity and diversity simultaneously, and to start to understand the quantity per group.

For the object grid views associated with shape, use, and color groupings (referred to as “All” in the project), vessels were arranged according to similarity rankings. In the shape-based clusters, objects were ordered by distance to the cluster centroid so that the most representative forms appeared first. In the use and color categories, I selected representative objects manually to display on the “shelves” and thus ordered surrounding objects according to their similarity to those objects. This arrangement allowed each group to visually transition outward from its most characteristic forms toward more unusual or ambiguous examples.

The project also incorporates more traditional visualization techniques alongside the experimental silhouette views. Geographic distributions were shown through an interactive map built with Mapbox, while temporal distributions were displayed using a bucketed timeline spanning the full range of the dataset. Together, these visualizations allowed viewers to compare how similar forms, uses, and colors recur across geography and historical time.

Conclusion

We tend to think about human history as a process of becoming more advanced, more evolved, and more technologically sophisticated. Especially now, progress can feel like something that separates us from one another — who is more modern, more developed, more civilized. But pottery tells a different story.

Across thousands of years, across cultures and political boundaries, humans have continued to make remarkably similar objects to meet remarkably similar needs. We still need to eat, drink, store water, gather in community, and create beauty in our everyday lives. Despite enormous changes in technology and society, many of the forms we use to do those things have remained surprisingly constant.

We're also living in an increasingly digital and abstract world — one where physicality itself can sometimes feel secondary and counter-cultural. Pottery reminds us that we are still physical beings living in a physical world - we still pick up and hold objects; weight, balance, texture, and form still matter to us.

Perhaps most interestingly, pottery has persisted even when our practical need for it has waned. Over the last century we've developed countless new materials that are cheaper, lighter, faster to produce, and often more durable than ceramics. In many ways, pottery should have become obsolete, but it hasn't. We still choose ceramic mugs over plastic ones. We still place flowers in ceramic vases and we still eat from bowls and plates made from fired earth. We still make pottery by hand as hobbyists, even when machines could do it more efficiently. This persistence suggests that these objects continue to resonate with something fundamentally human.

The goal of this project was never to erase the cultural, historical, or artistic differences between these vessels. Those differences are meaningful, beautiful, and important. Instead, by reducing the vessels to their silhouettes and focusing on form, I hoped to create a moment where viewers could pause before immediately categorizing the objects by origin, style, or time period, and instead notice the similarities first.

There are an uncountable number of ways to shape a vessel, and yet similar forms continue to emerge again and again across centuries and across continents. The persistence of those forms suggests that beneath enormous cultural and technological change, there are still fundamental aspects of human life that remain remarkably constant.

The practice of pottery shows us how connected we really are — not only to people living around us today, but also to those who came thousands of years before us.

Appendix

Project Images

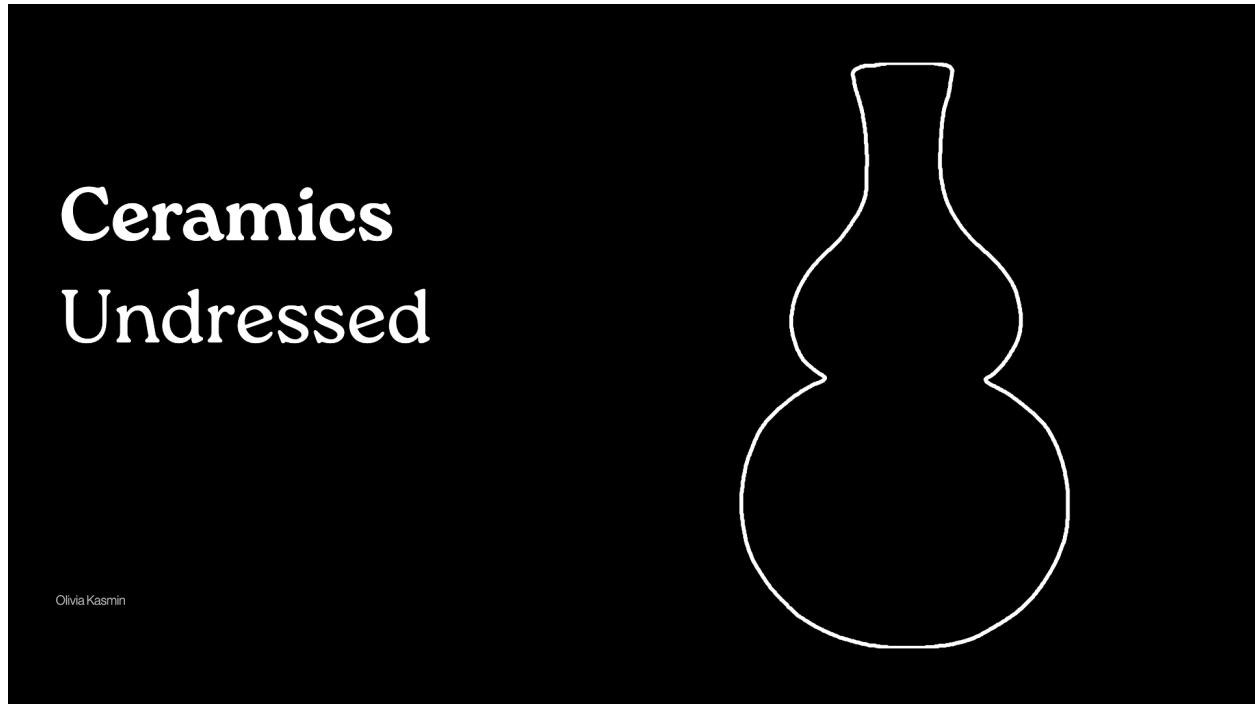


Figure A-1: title



We've been making **ceramic vessels** for over 20,000 years.

When we **undress** them a shared thread emerges.



Figure A-2: intro

Anatomy of a vessel



neck

belly

ceramic *(ncl)*

"made from clay that has been shaped and then baked until hard"

vessel *(v)*

"a container (such as a cask, bottle, kettle, cup, or bowl) for holding something"

Figure A-3: anatomy of a vessel



Figure A-4: data intro

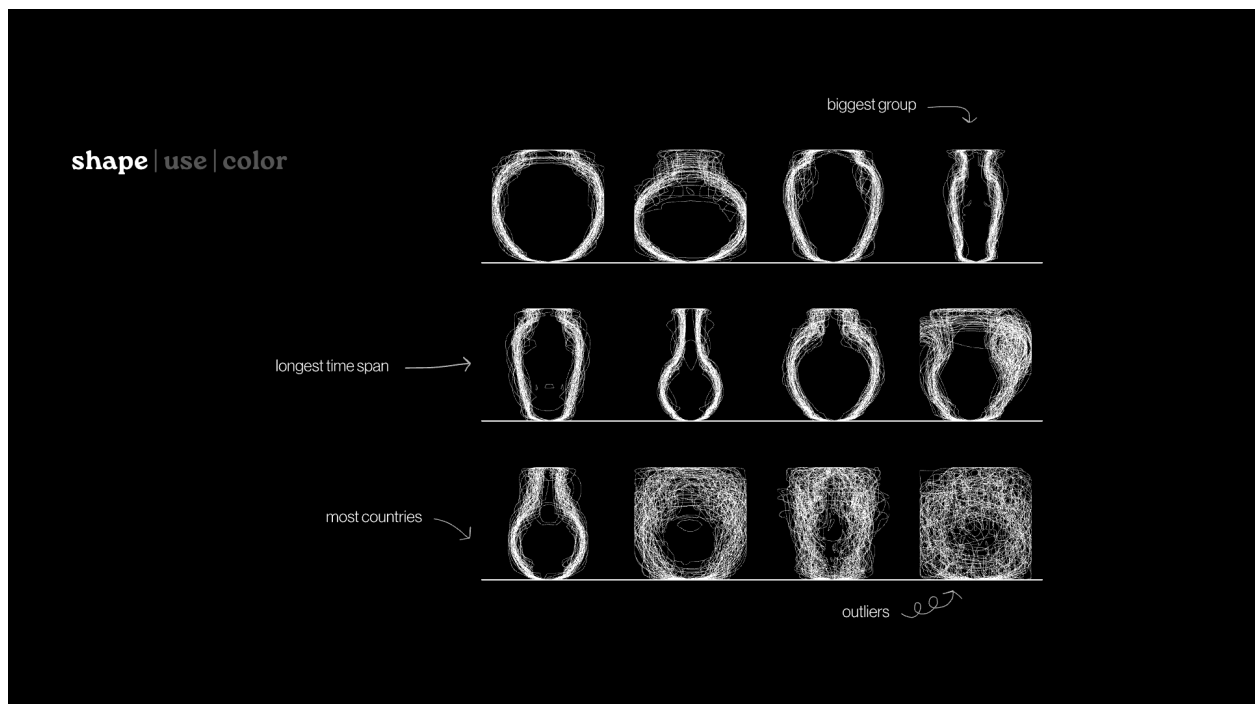


Figure A-5: shelf view for shape clusters

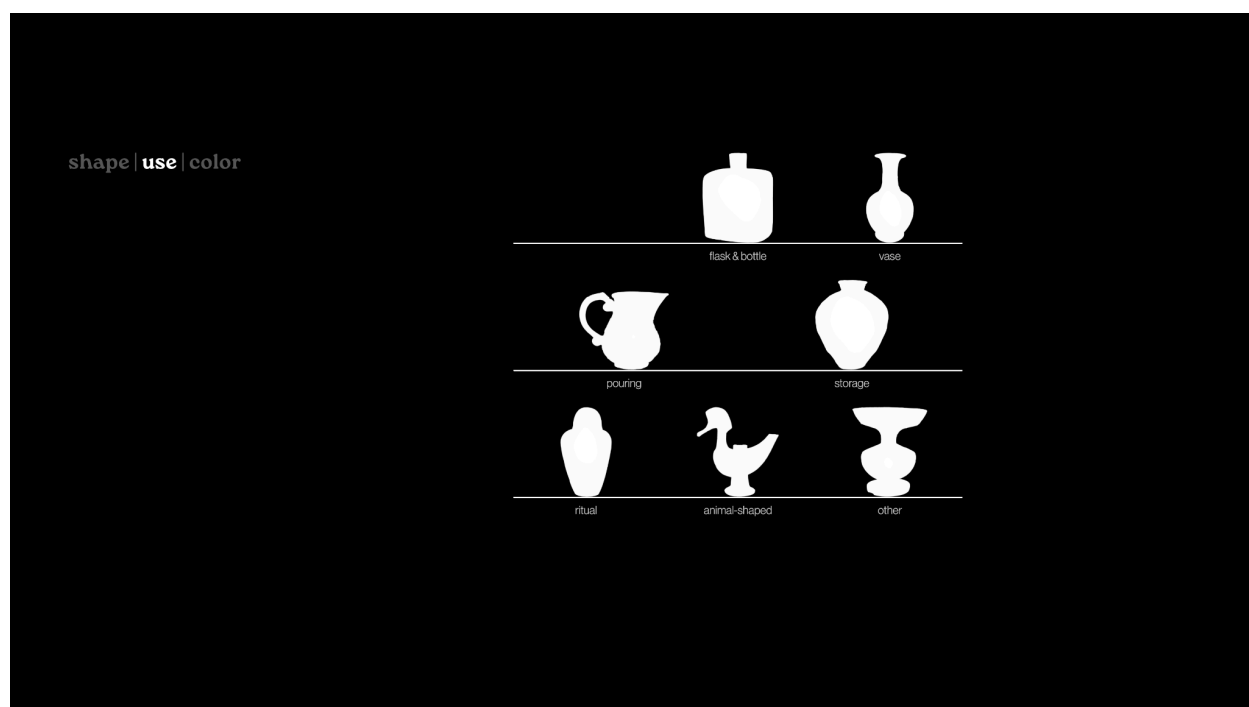


Figure A-6: shelf view for use groups

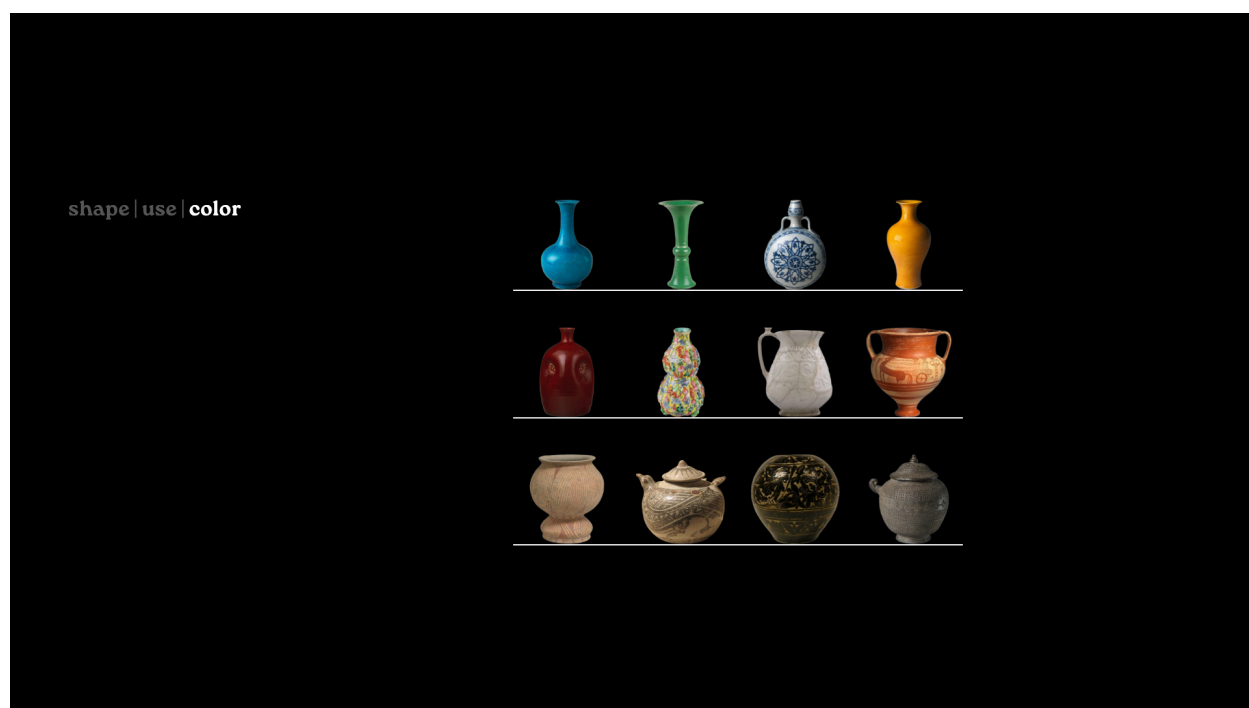


Figure A-7: shelf view for color groups

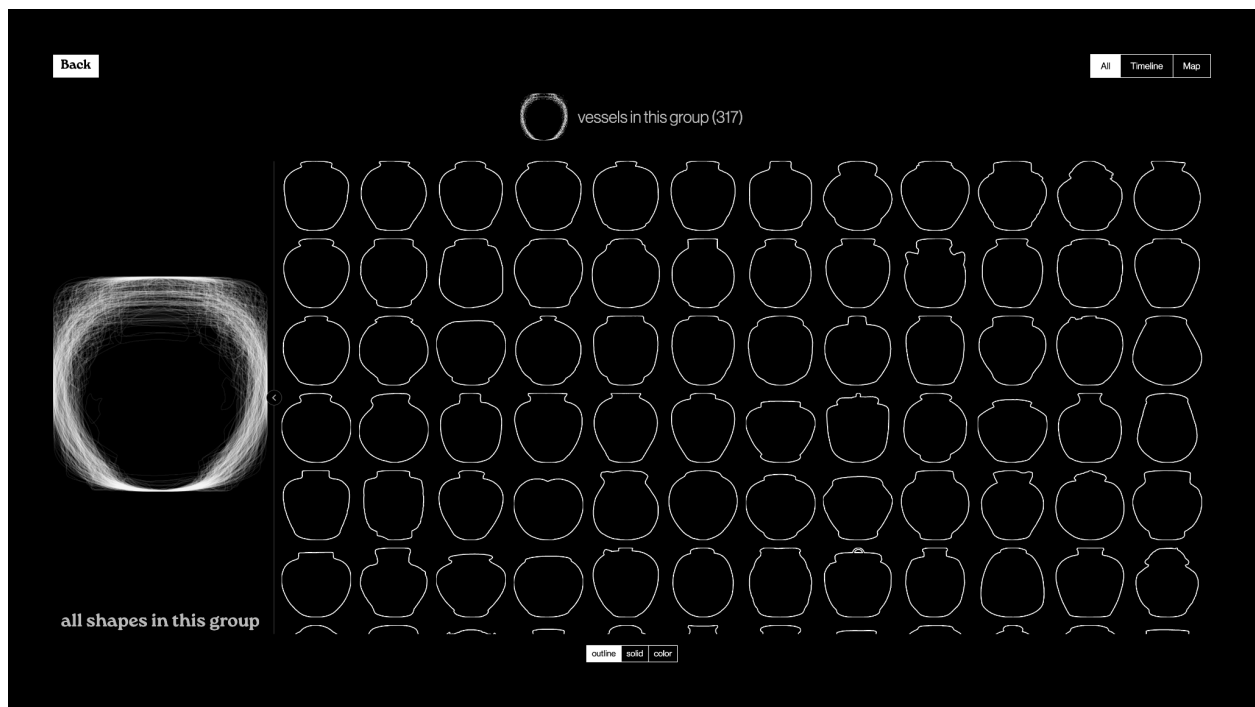


Figure A-8: All view for a single shape cluster

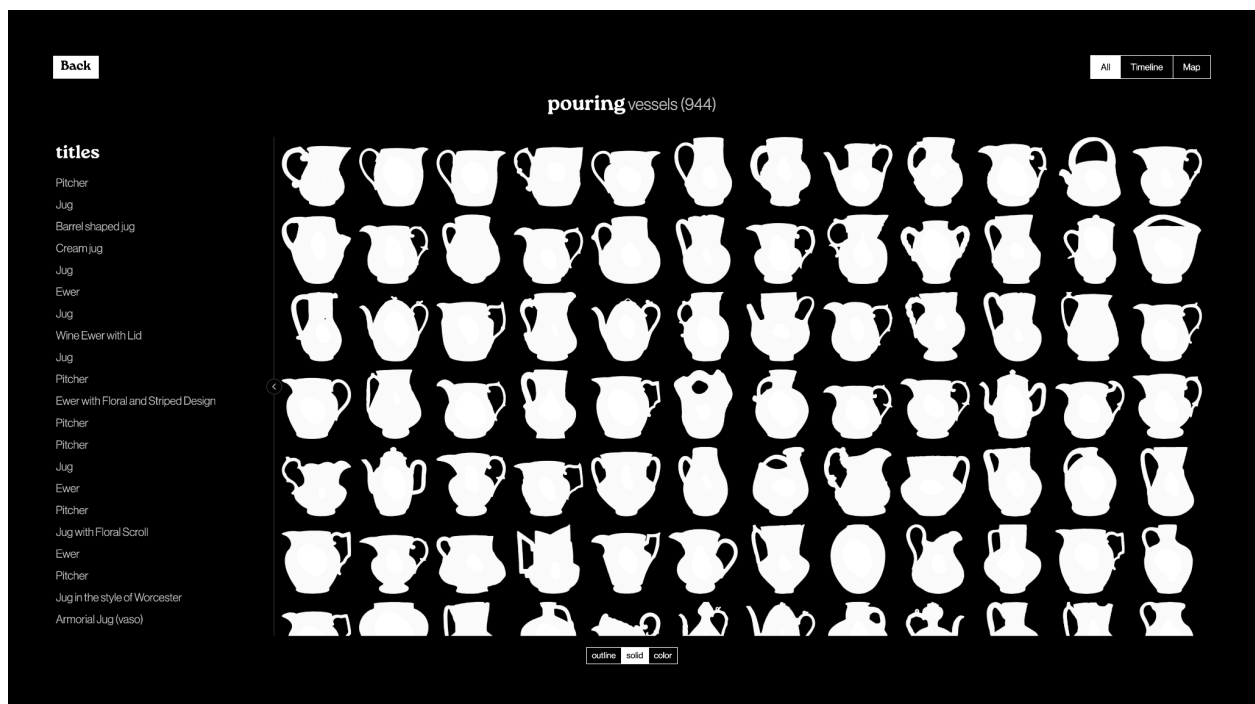


Figure A-9: All view for a single use group



Figure A-10: All view for a single color group

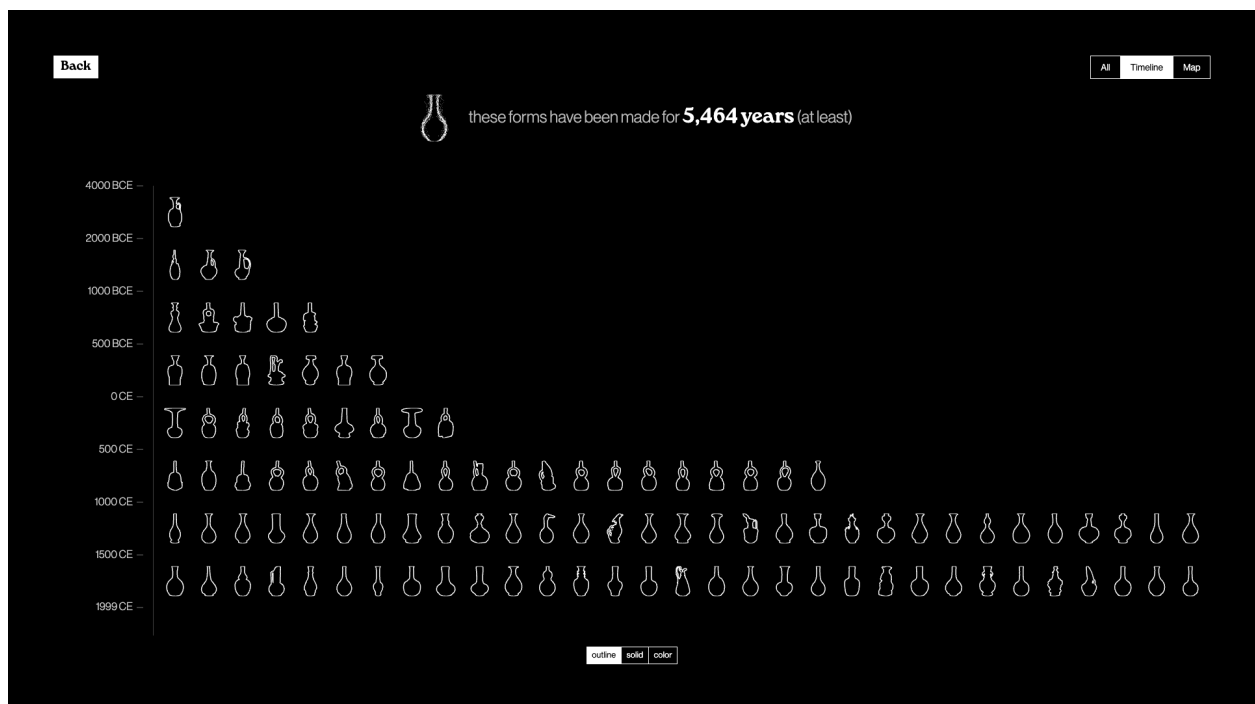


Figure A-11: Timeline view for a single shape cluster

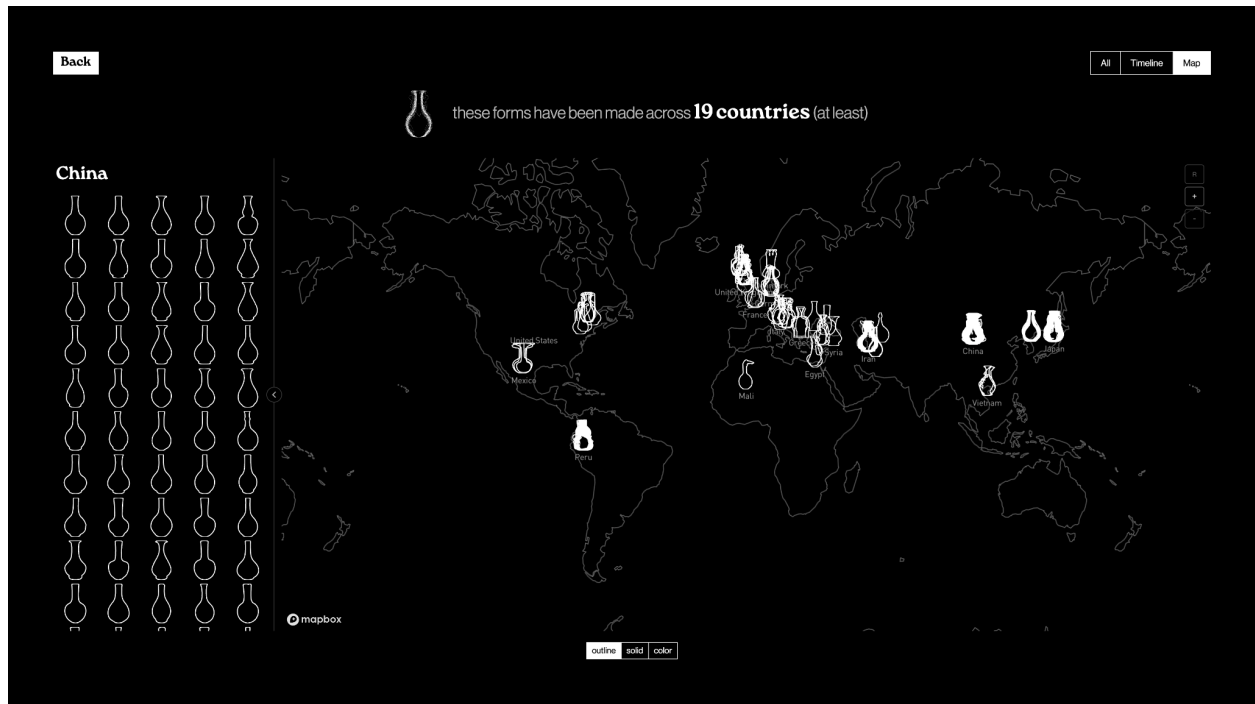


Figure A-12: Map view for a single shape cluster

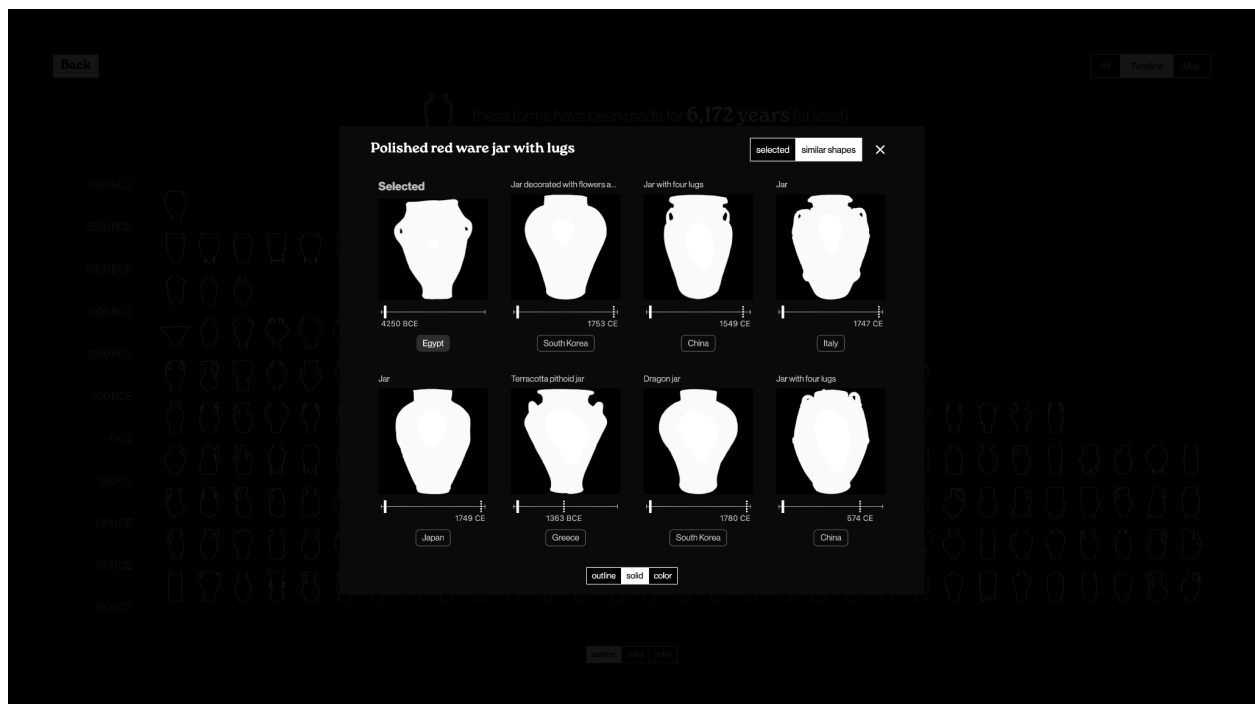


Figure A-13: nearest neighbors modal

Same silhouette, thousands of years, different cultures.



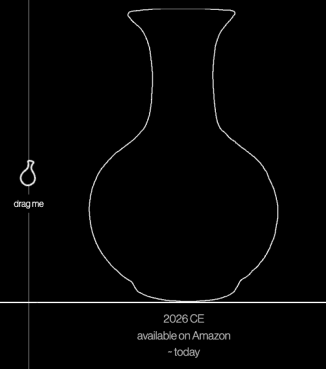
850 BCE–750 BCE
Cyprus
~ 2576 years ago



600 CE–699 CE
Japan
~ 1426 years ago



1700 CE–1799 CE
China
~ 326 years ago



2026 CE
available on Amazon
~ today

Figure A-14: case studies conclusion

Sketches

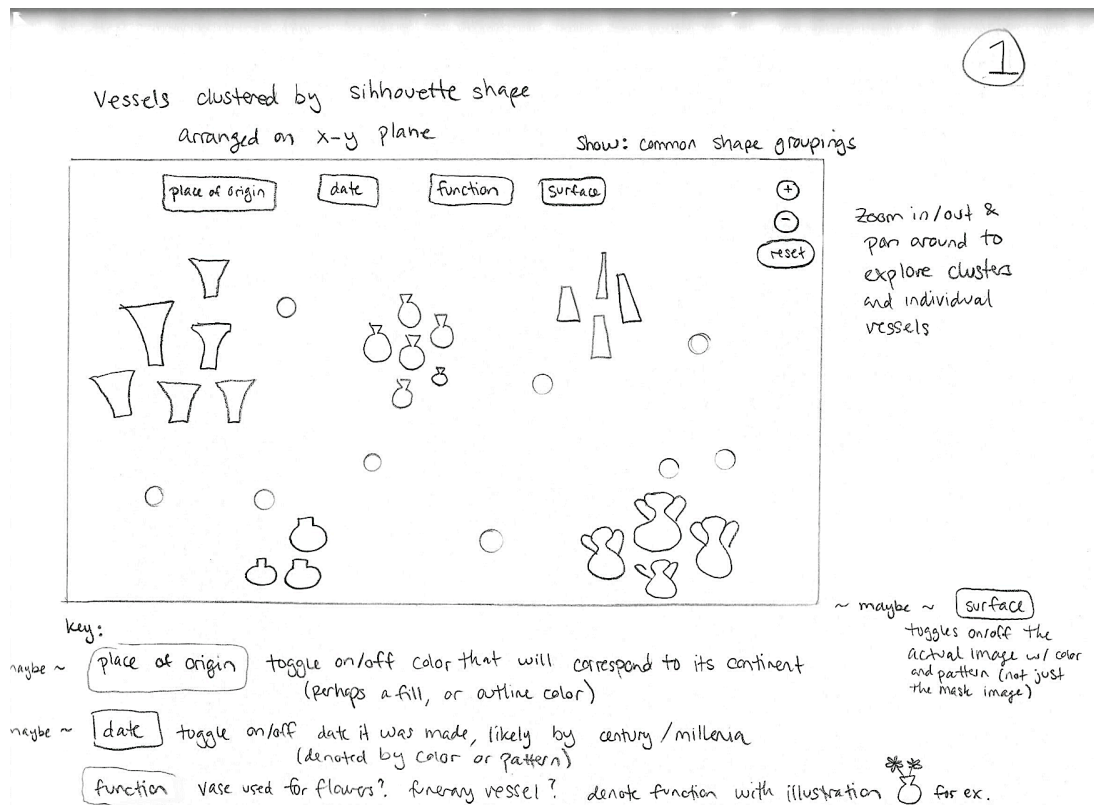


Figure A-15: sketch for shape cluster display

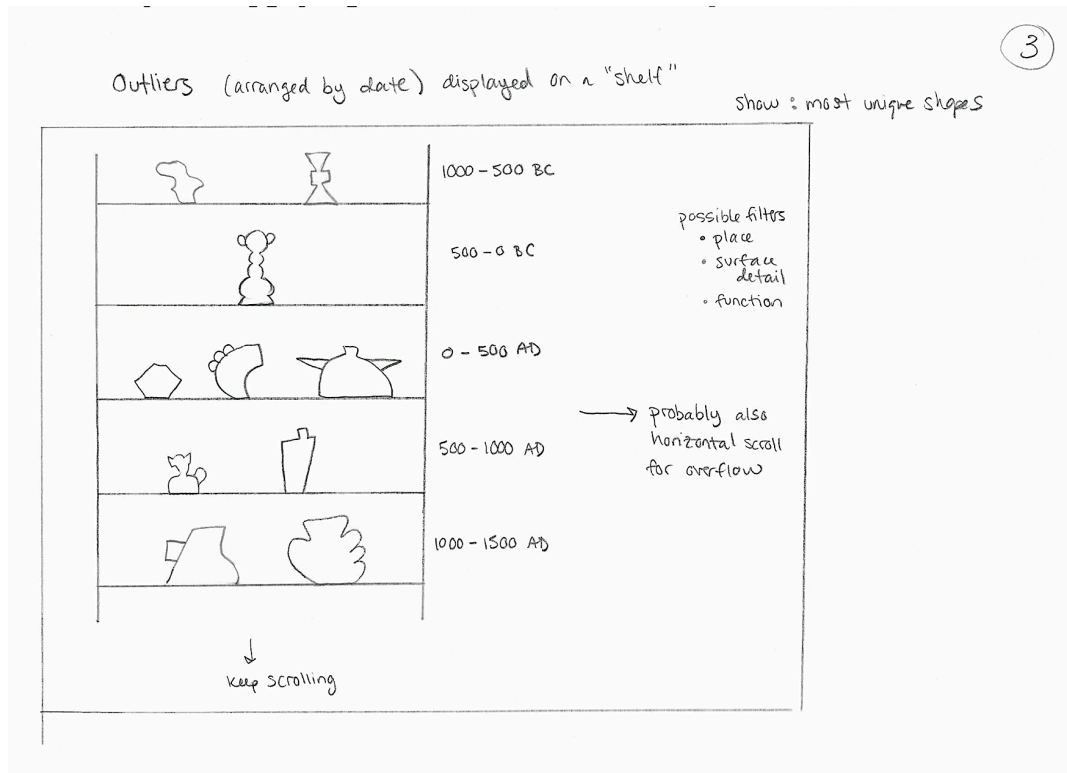


Figure A-16: sketch for outlier shapes shelf

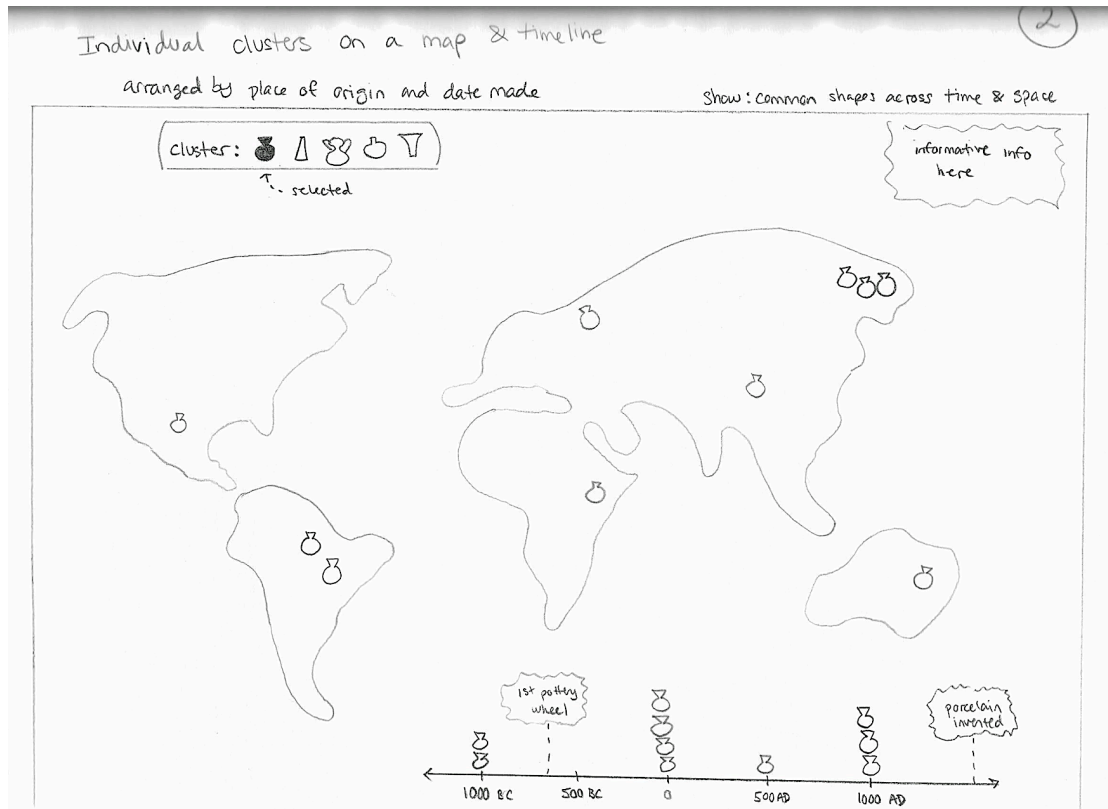
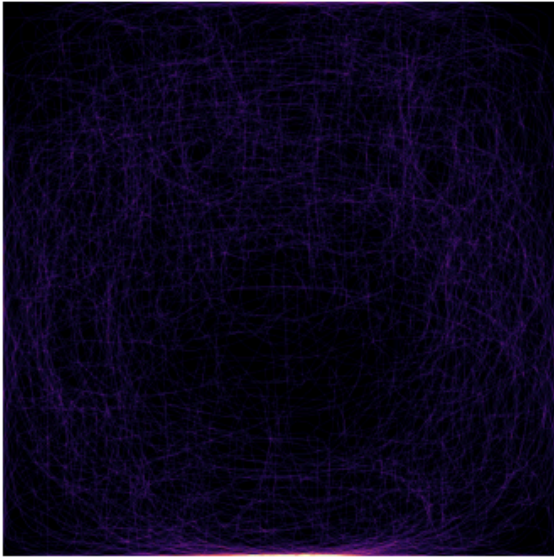


Figure A-17: sketch for single shape cluster on map and timeline

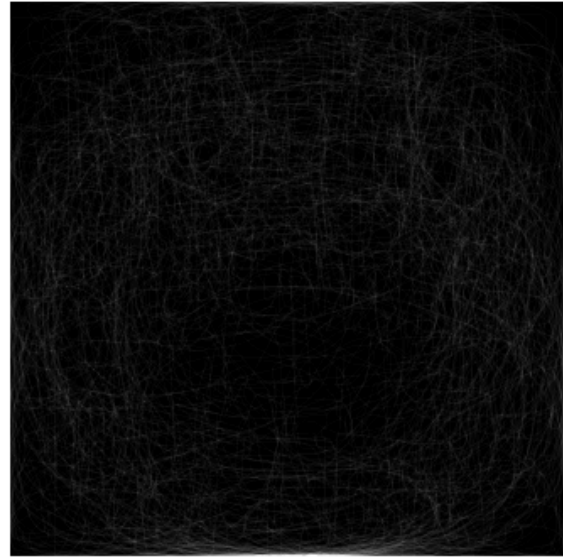
Outline Shape Cluster Heatmaps

Stacked Outlines - Cluster 0

Heatmap (n=126)

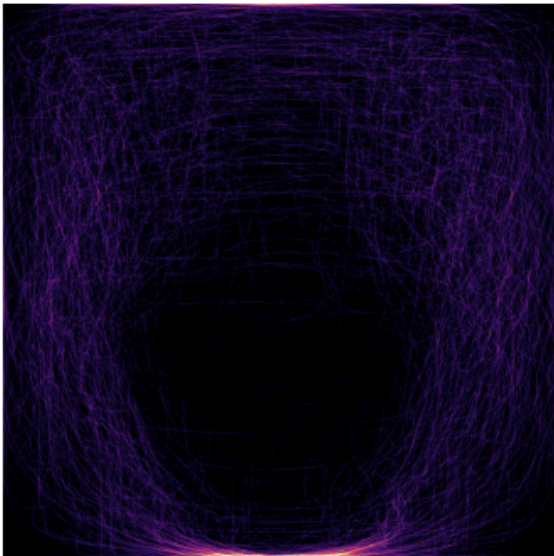


White Overlay (n=126)

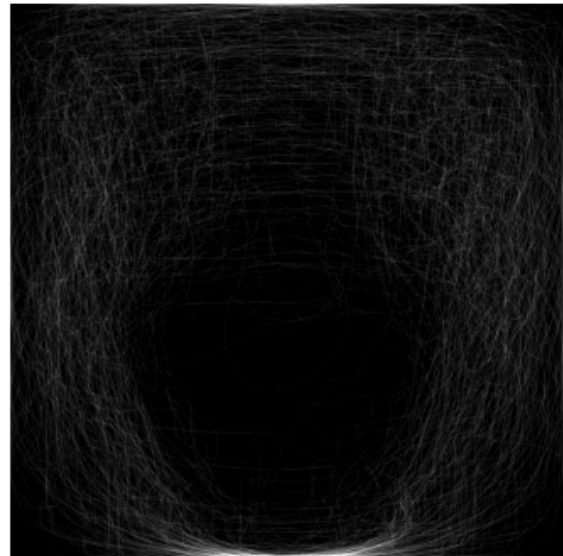


Stacked Outlines - Cluster 1

Heatmap (n=171)

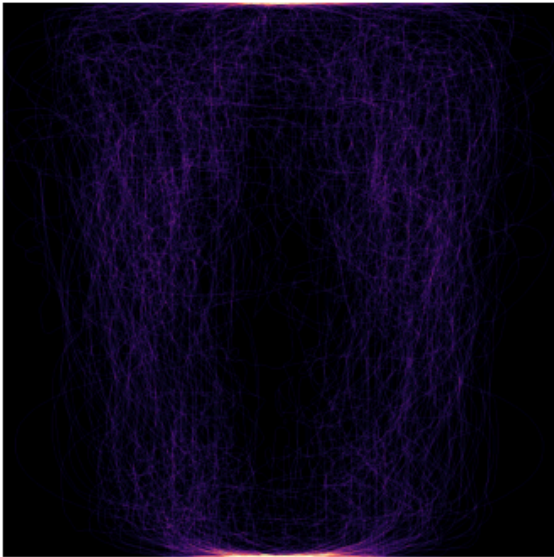


White Overlay (n=171)

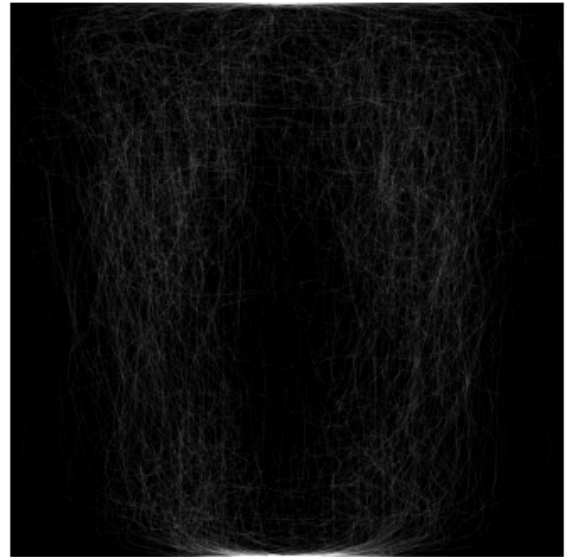


Stacked Outlines - Cluster 2

Heatmap (n=122)

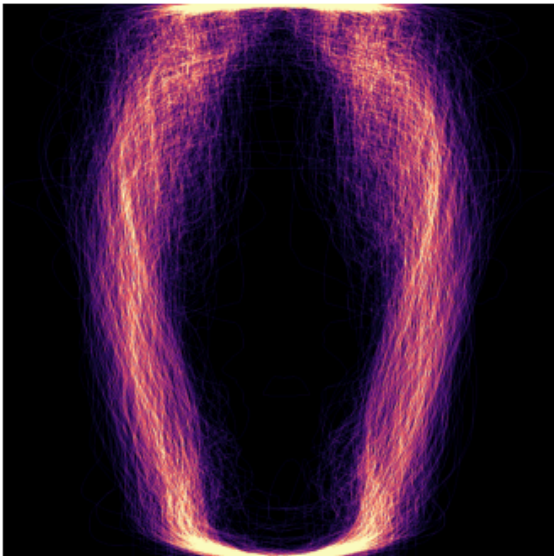


White Overlay (n=122)

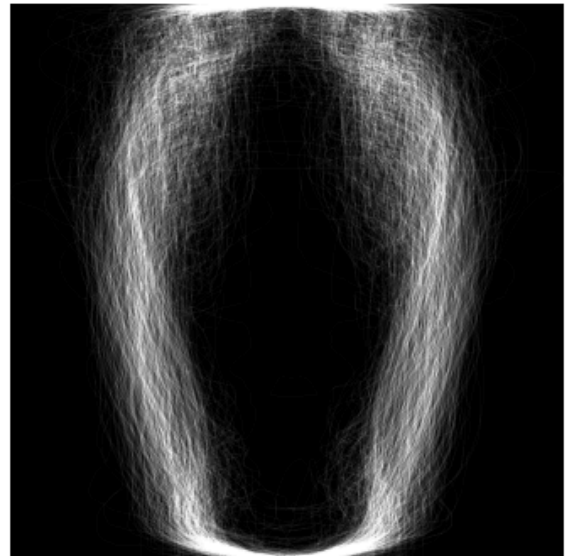


Stacked Outlines - Cluster 3

Heatmap (n=634)

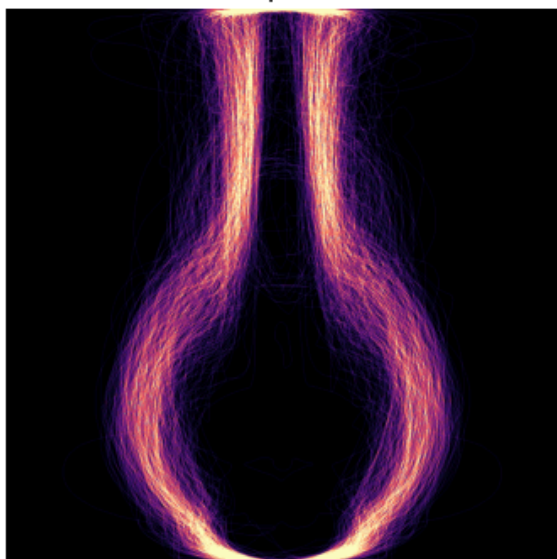


White Overlay (n=634)

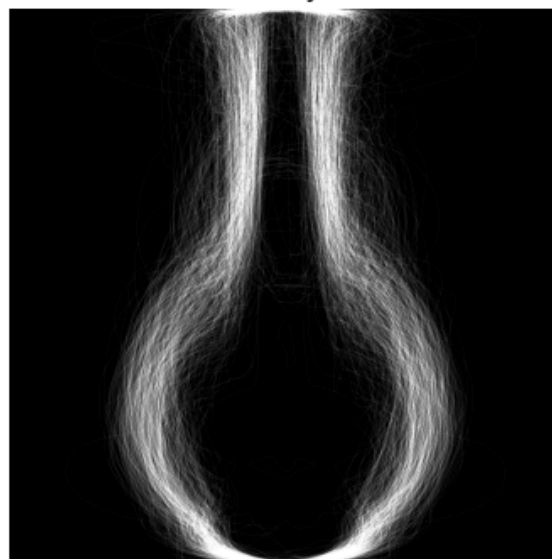


Stacked Outlines - Cluster 4

Heatmap (n=528)

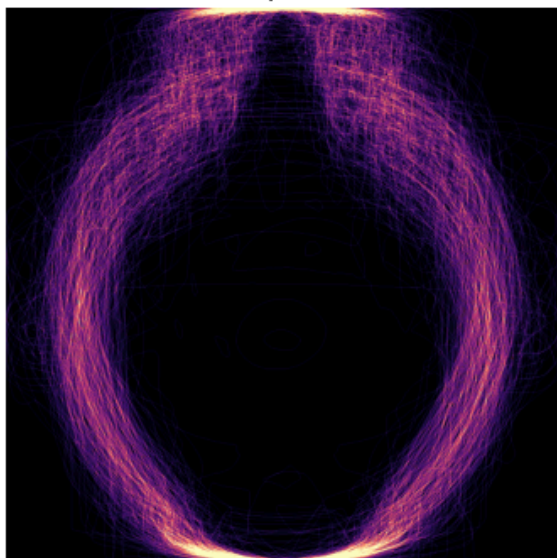


White Overlay (n=528)

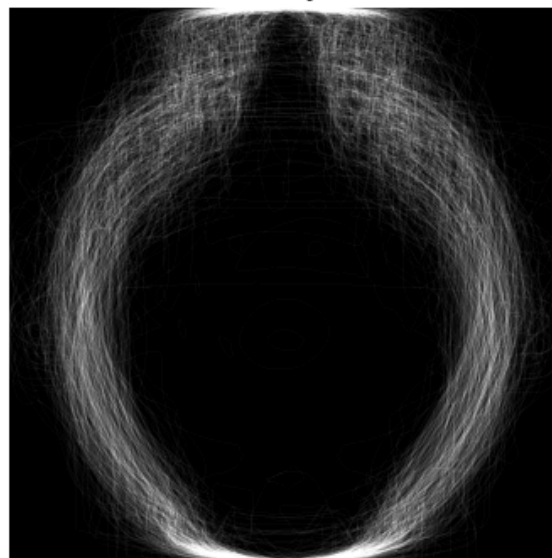


Stacked Outlines - Cluster 5

Heatmap (n=413)

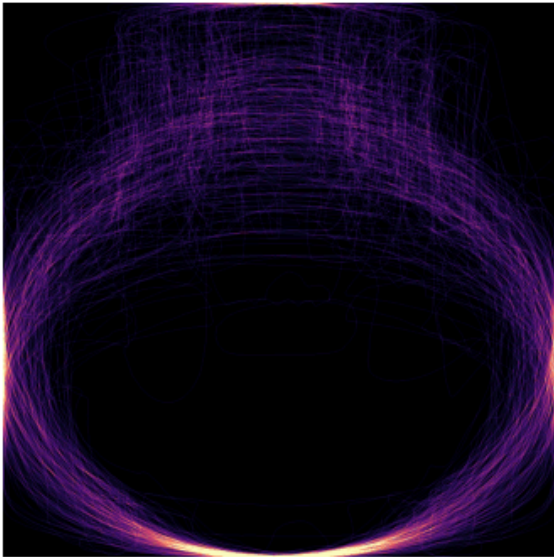


White Overlay (n=413)

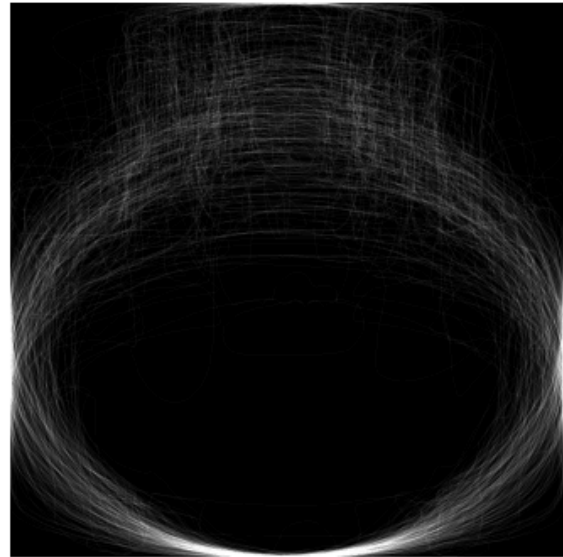


Stacked Outlines - Cluster 6

Heatmap (n=232)

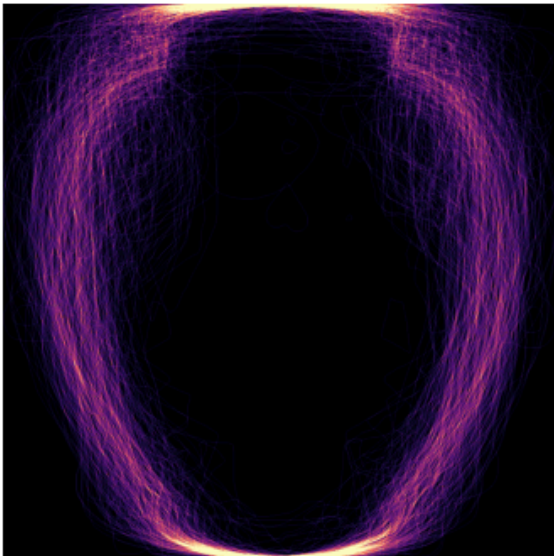


White Overlay (n=232)

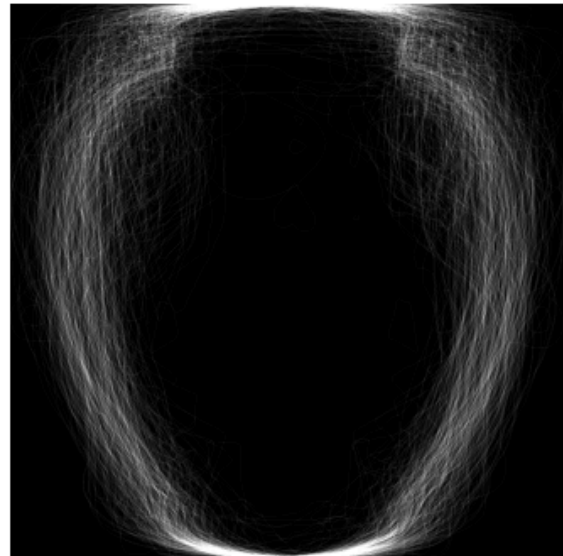


Stacked Outlines - Cluster 7

Heatmap (n=338)

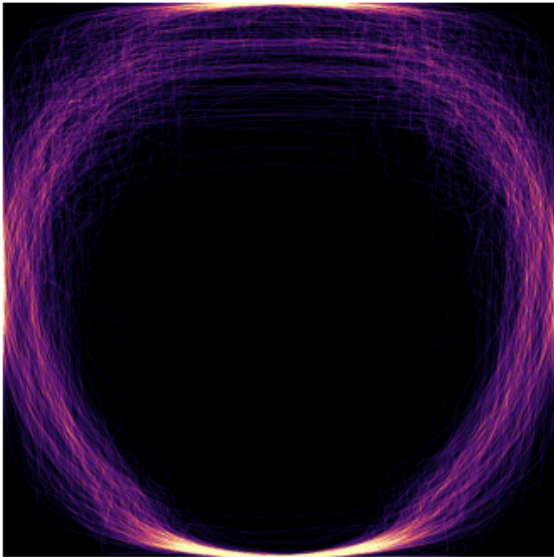


White Overlay (n=338)

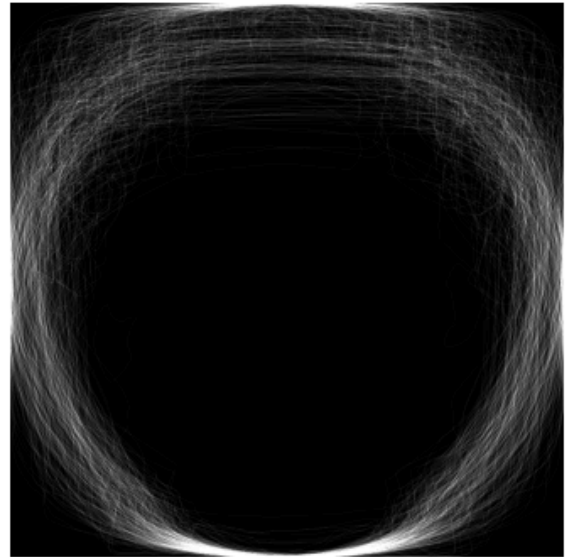


Stacked Outlines - Cluster 8

Heatmap (n=317)

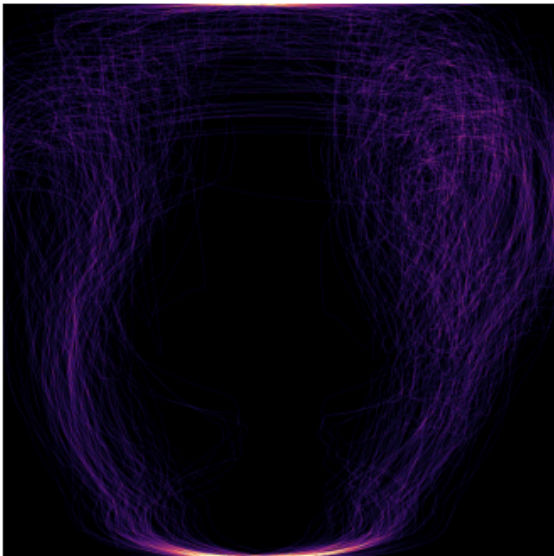


White Overlay (n=317)

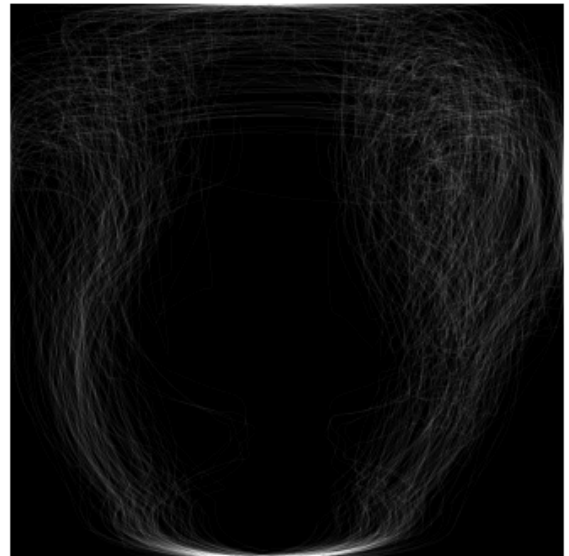


Stacked Outlines - Cluster 9

Heatmap (n=133)

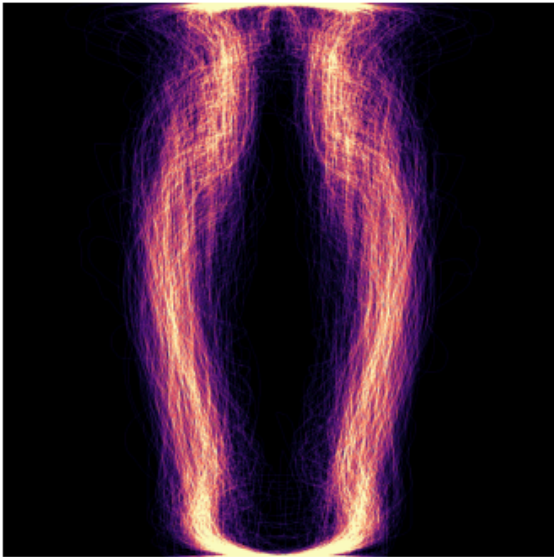


White Overlay (n=133)

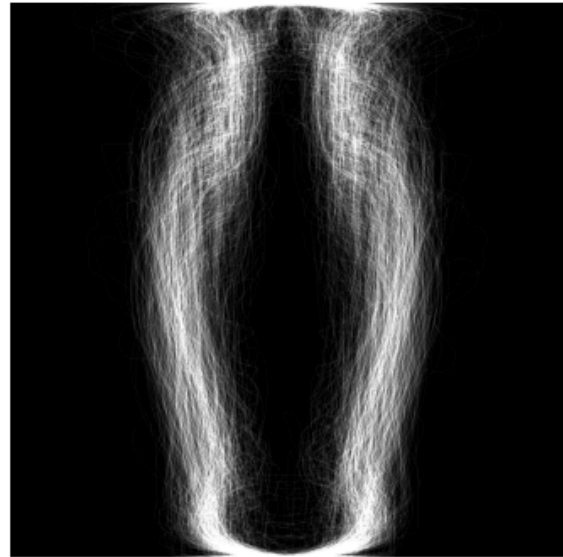


Stacked Outlines - Cluster 10

Heatmap (n=685)

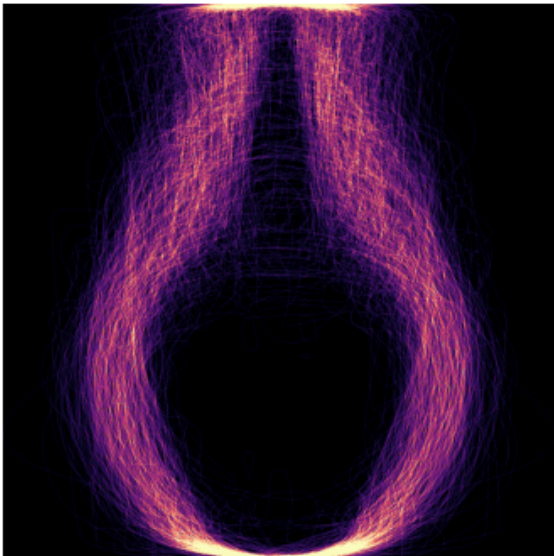


White Overlay (n=685)

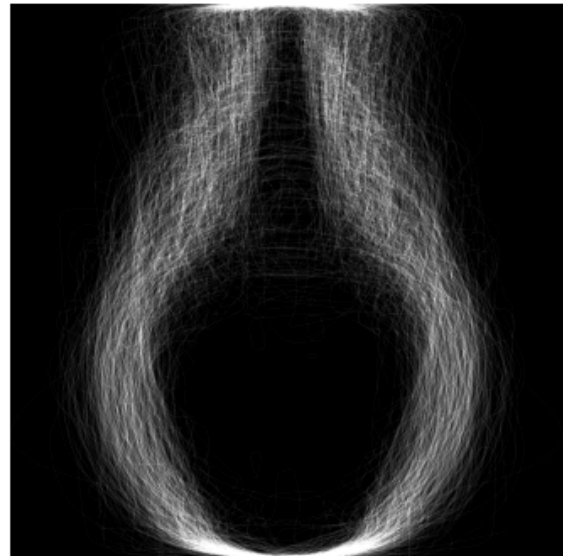


Stacked Outlines - Cluster 11

Heatmap (n=487)



White Overlay (n=487)



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